Computer Networks

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Unit - 02: Physical Layer & Network Media

Class Load : 4 hrs

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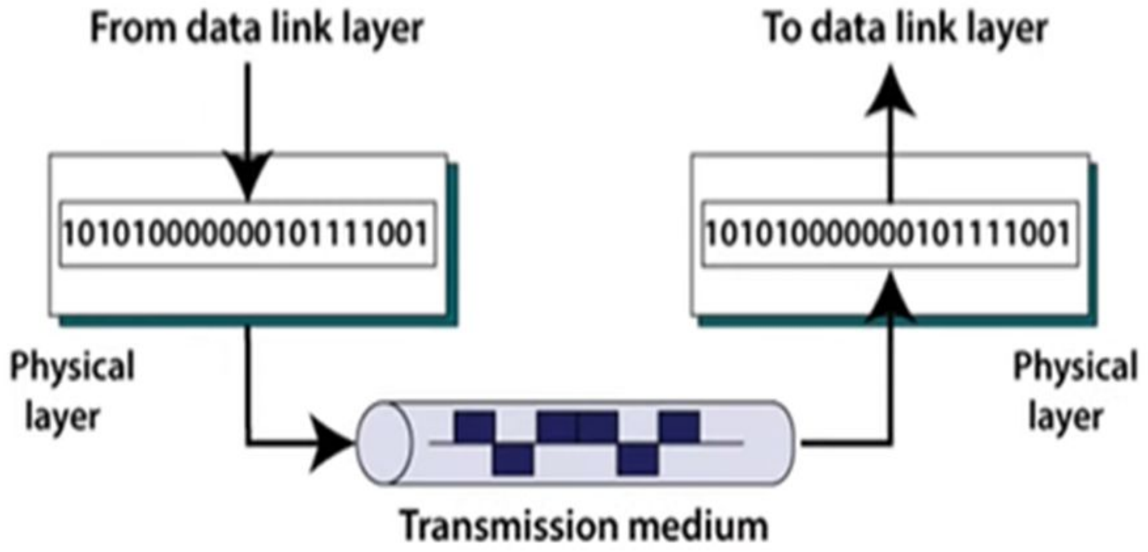
Physical Layer (Revisit):

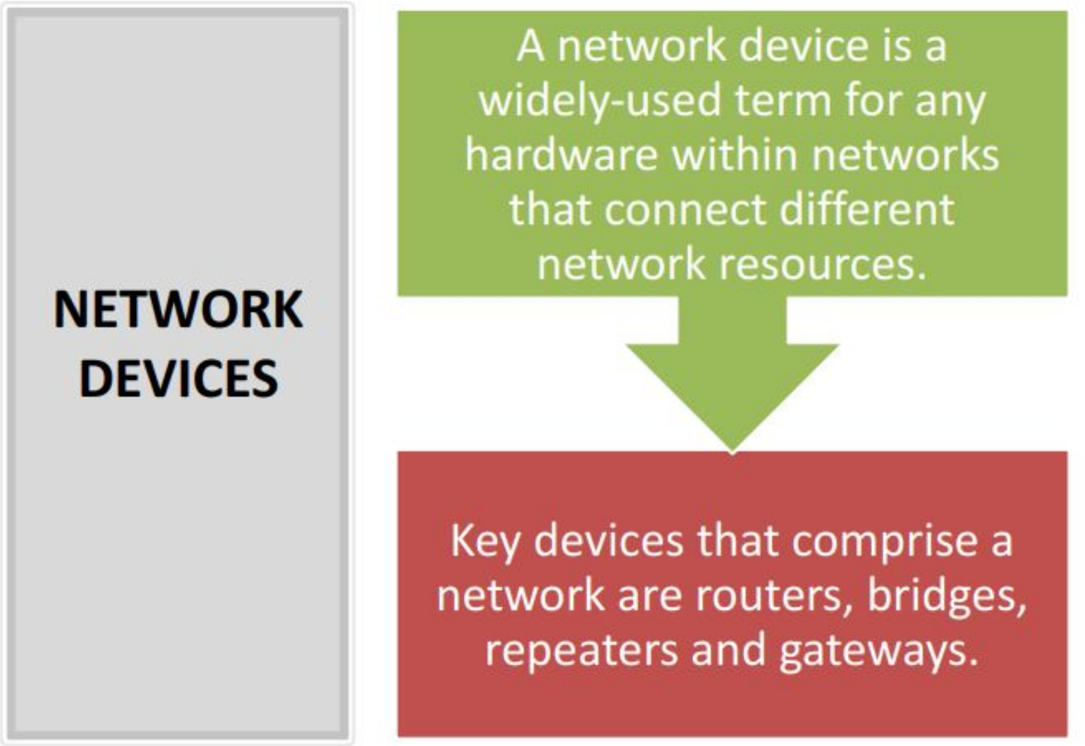
- In the OSI model, the physical layer is the lowest layer, and it deals with the actual physical transmission of data over a physical medium, such as copper wires, optical fibres, or wireless radio waves.
- It defines the hardware characteristics and electrical/optical specifications of the transmission medium (i.e voltage, data rates etc).
- Examples of devices at the OSI physical layer include Hub, Repeater, Modem, Cables.
- It deals with the mechanical and electrical specifications of the interface and transmission medium. It converts the digital/analog bits into electrical signal or optical signals.
- It activates, maintains and deactivates the physical connection.
- Data encoding is also done in this layer.
- **PDU : Bits**

Physical Layer (Revisit):



Network Devices 2.1

Definition:


NETWORK DEVICES
A network device is a widely-used term for any hardware within networks that connect different network resources.
Key devices that comprise a network are routers, bridges, repeaters and gateways.

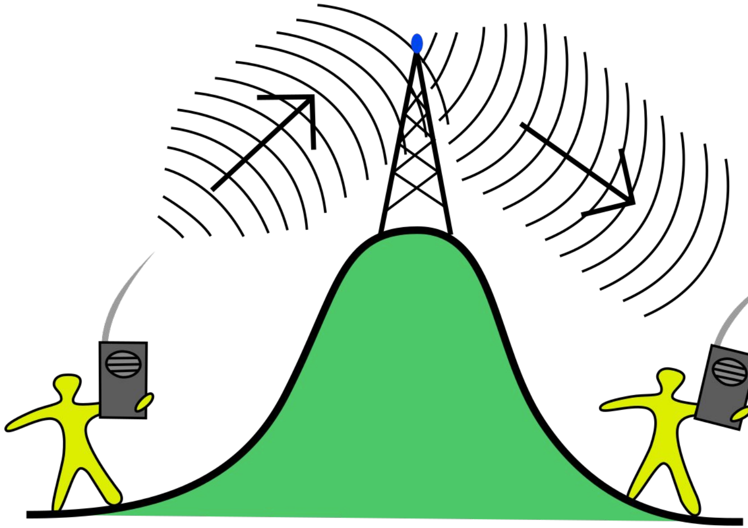
Repeater:

A repeater is one of the computer networking devices that operates at the **physical layer** of the OSI model.

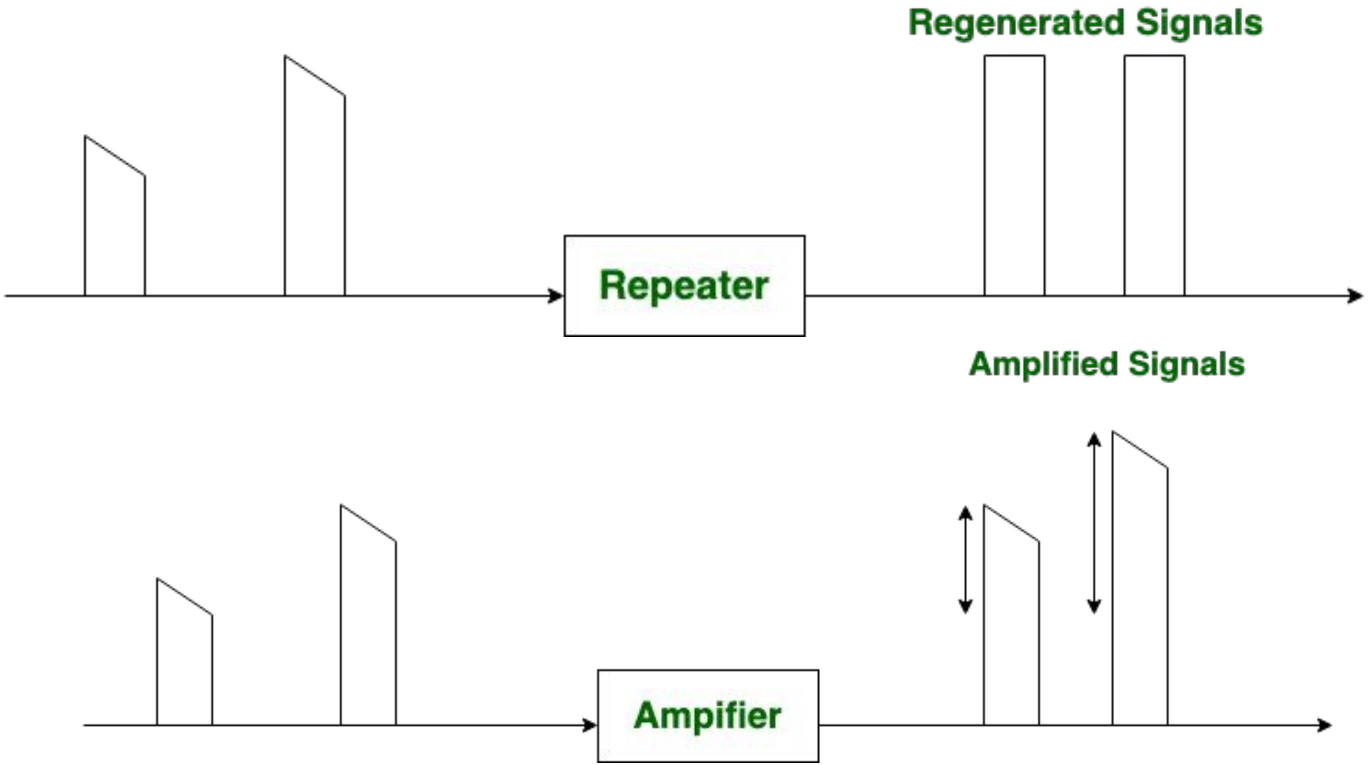
- Its job is to **regenerate the signal** over the same network before the signal becomes too weak or corrupted so as to **extend the length** to which the signal can be transmitted over the same network. (They **copy** the signal **bit by bit** and regenerate it at the original strength.)
- **Primary Function:**
  - **Regenerate** signals and **do not amplify** signals. As data travels over a network, the signal strength tends to weaken over long distances. The repeater boosts the signal to maintain its integrity.
- Repeaters require a **small amount of time** to regenerate the signal which can cause propagation **delay** affecting communication when there are several repeaters connected in a row.

Repeater:


Wi Fi
REPEATER
Power
Wireless
WPS
LAN
US Plug
LAN
LAN Port
WPS
WPS
Button
Reset
Reset Button



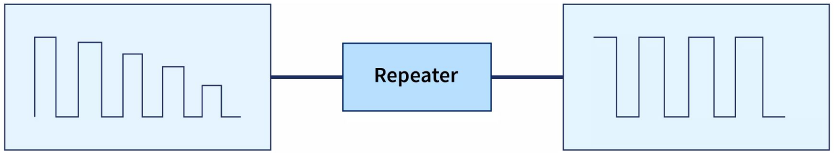
Repeater:

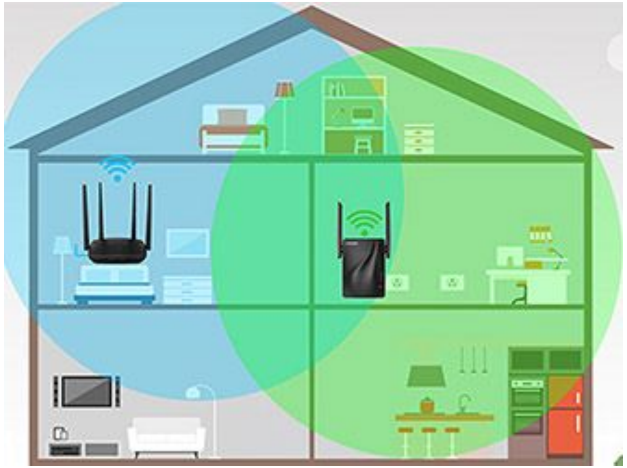


Regenerated Signals
Repeater
Amplified Signals
Ampifier

Repeater:

- Many network architectures limit the number of repeaters that can be used in a row.
- Repeaters work only at the physical layer of the OSI network model.
- Normally it used in wireless communication, they may be used to extend the coverage area of Wi-Fi networks. (Used in both Wired and Wireless Networks).





Repeater:

Advantages of Repeater:

- Repeaters can increase the overall distance of a network.
- Repeaters are easy to set up and can easily increase network length or coverage area.
- Repeaters have no significant impact on network performance.
- It is a cost-effective network device.

Disadvantages of Repeater:

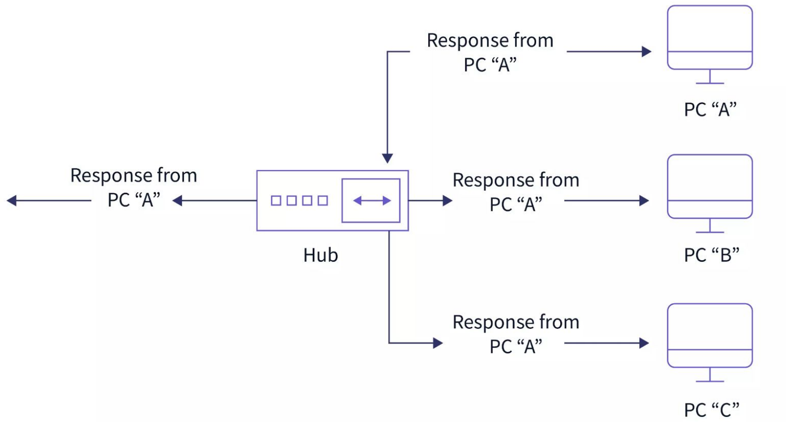
- Repeaters are unable to connect disparate networks.
- Repeaters cannot reduce network traffic.
- Most repeaters on a network generate noise on the wire, increasing the possibility of packet collisions.

Hub:

A hub is a physical-layer device that acts on **individual bits**. When a bit, representing a **zero or a one**, arrives from one interface, the hub simply **recreates the bit**, **boosts** its energy strength, and **transmits the bit** into **all** the other interfaces.

- A hub is a **repeater with multiple port.**
- A hub is a **central device** or system that connects **multiple devices** or networks.
- A hub is often a **platform or device** that serves as a central point for accessing and distributing various types of content, such as articles, videos, or podcasts.
- It can be used to **create multiple** levels of **hierarchy** of stations.
- The stations connect to the hub with RJ-45 connector having maximum segment length is 100 meters.

Hub:


Response from
PC "A"
PC "A"
Response from
PC "A"
Hub
Response from
PC "A"
PC "B"
Response from
PC "A"
PC "C"


IBM

Hub:

Active Hub:

- These hubs have their **power source** and can **clean, enhance, and relay** the network's signal.
- It functions as both a **repeater** and a **wiring center**.
- The active hub may **repair damaged packets** as they are being sent and can also **hold** the direction of the remaining packets and distribute them.

Passive Hub:

- This does not use electricity, they are more like **point contact** for the **wires** to built in the physical network.
- These hubs **do not** clean or enhance signals before relaying them to the network and cannot be utilized to extend the distance between nodes.

Hub:

Advantages of Hub:

- Due to its low cost, anyone can utilize it.
- It supports various types of Network Media.
- The presence of a hub does not affect network performance.

Disadvantages of Hub:

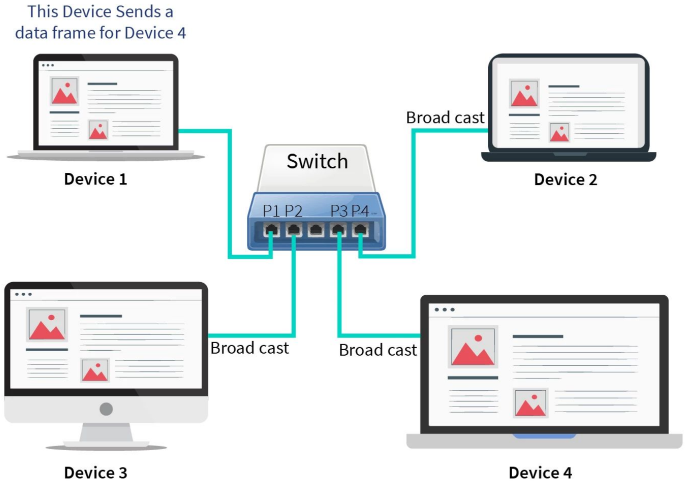
- It is unable to select the best network path.
- It has no mechanism for reducing network traffic.
- It is unable to filter information because it sends packets to all connected segments.

Switch:

A switch is networking devices that connects **multiple devices** (computers , printers, etc.) together on LAN and **forwards** data between them based on **MAC address.**

- Switch is ***data link layer*** device.
- A switch is a **multiport bridge** with a buffer and a design that can boost its efficiency (large number of ports imply less traffic) and performance.
- Switch can perform **error checking** before forwarding data, that makes it very efficient as it **does not** forward packets **that have** errors and forward good packets selectively to **correct port** only.
- Switches are **more advanced** than bridges and are designed to connect multiple devices within the same LAN.

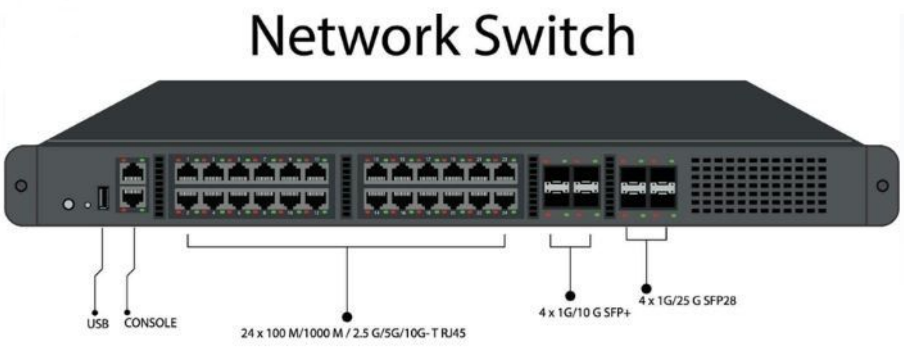
Switch:



Post	MAC
1	-
2	-
3	-
4	MAC of device 4

Steps :-

1 Device 1 send data frame for device 4 .
2 Since MAC table is empty .
3 Switch broadcast to all the device except 1.
4 Since the data frame was for device 4, it sends an acknowledgement.
5 Now switch maps post 4 to device 4 .



Approach of Forwarding Packets in Switch:

- **Cut-through:**
  - A switch forwards a frame **immediately** after receiving the destination address. As a consequence, the switch forwards the frame **without** collision and error detection.
- **Collision-free:**
  - In this case, the switch forwards the frame after receiving **64 bytes**, which allows detection of collision. However, error detection is not possible because switch is yet to receive the entire frame.
- **Fully buffered:**
  - In this case, the switch forwards the frame only after receiving the **entire** frame. So, the switch can detect both collision and error free frames are forwarded.

Types of Switch:

There are generally three types of switches.

➔ **Managed Switch :**

- These are **expensive** switches used in organizations with large and complex networks because these types of switches can be customized to augment the functionalities of a standard switch.
- Manageable switch has a console port and IP address, which can be assigned and configured.

➔ **Unmanaged Switch :**

- These are low-cost switches that are commonly found in home networks and small businesses.
- Configuration can't be made. It is not possible to assign IP address as there is no console port.

➔ **LAN Switch :**

- LAN switches connect devices on an organization's internal LAN. They are also known as Ethernet switches or data switches. These switches are especially useful for reducing network congestion or bottlenecks.

Switch:

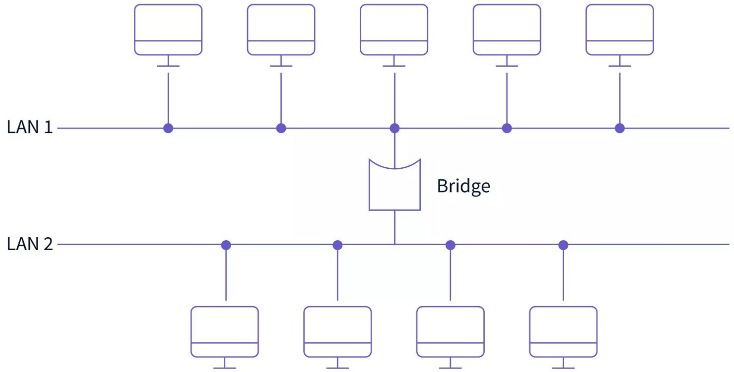
- **Advantages of Switch:**
  - Switches **boost** the network's available bandwidth.
  - Switch-based networks will have **fewer frame collisions** because switches create collision domains for each connection.
  - Switches improve the network's efficiency.

- **Disadvantages of Switch:**
  - It is challenging to identify network connectivity problems using the network switch.
  - Switches in **promiscuous mode** are vulnerable to security attacks such as IP spoofing and frame capture.

Bridge:

- A bridge operates at data link layer.
- Bridges are designed to **connect and filter** traffic between separate but interconnected LANs.
- Used to connect **two or more** network segment.
- A bridge is a repeater, with add on functionality of **filtering content** by reading the MAC addresses of source and destination.
- It is also used for interconnecting **two LANs** working on the same protocol.
- Some important Features of Bridges.
  - A bridge **operates both** in physical and data-link layer.
  - A bridge uses **a table** for filtering/routing.
  - A bridge **does not change** the physical (MAC) addresses in a frame.

▶ Bridge:



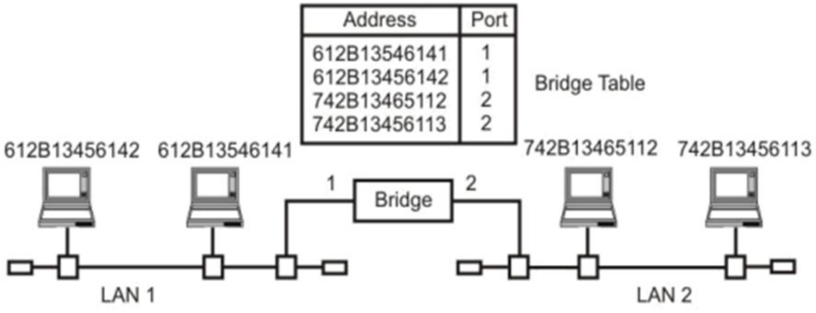


Figure A bridge connecting two separate LANs

Types of Bridge:

- **Transparent Bridges :**
    - These bridges connect different network parts **without the computers knowing** they're there. They learn where each computer is and send data only to the right part.
    - These bridges operate in a manner **that is transparent** to all networked hosts. A transparent bridge stores MAC addresses **in a table** similar to a routing table and **uses that information** to route packets to their destination.

- **Source Routing Bridges :**
    - The computer sending data **tells the bridge exactly** how to send it through the network. The bridge follows these directions to deliver the data.
    - The host can find the frame by sending a special frame known as the **discovery frame**, which propagates throughout the network **using all possible paths** to the destination.

Bridge:

Advantages of Bridge:

- Bridges reduce network traffic with minor segmentation.
- Bridges can also help to reduce network traffic on a segment by splitting up network communications.
- Bridge expands the number of connected workstations and network segments.
- It reduces collisions.

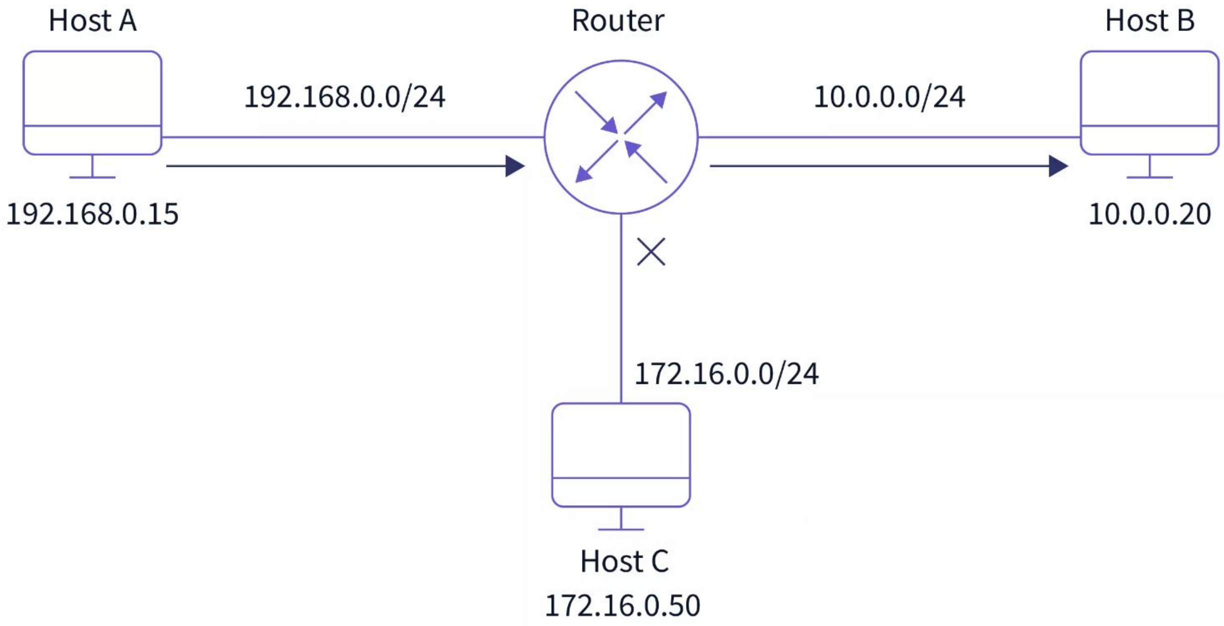
Disadvantages of Bridge:

- It is slower than repeaters due to the filtering process.
- A bridge is more expensive than repeaters or hubs.
- Bridges are not scalable to an extremely large network.

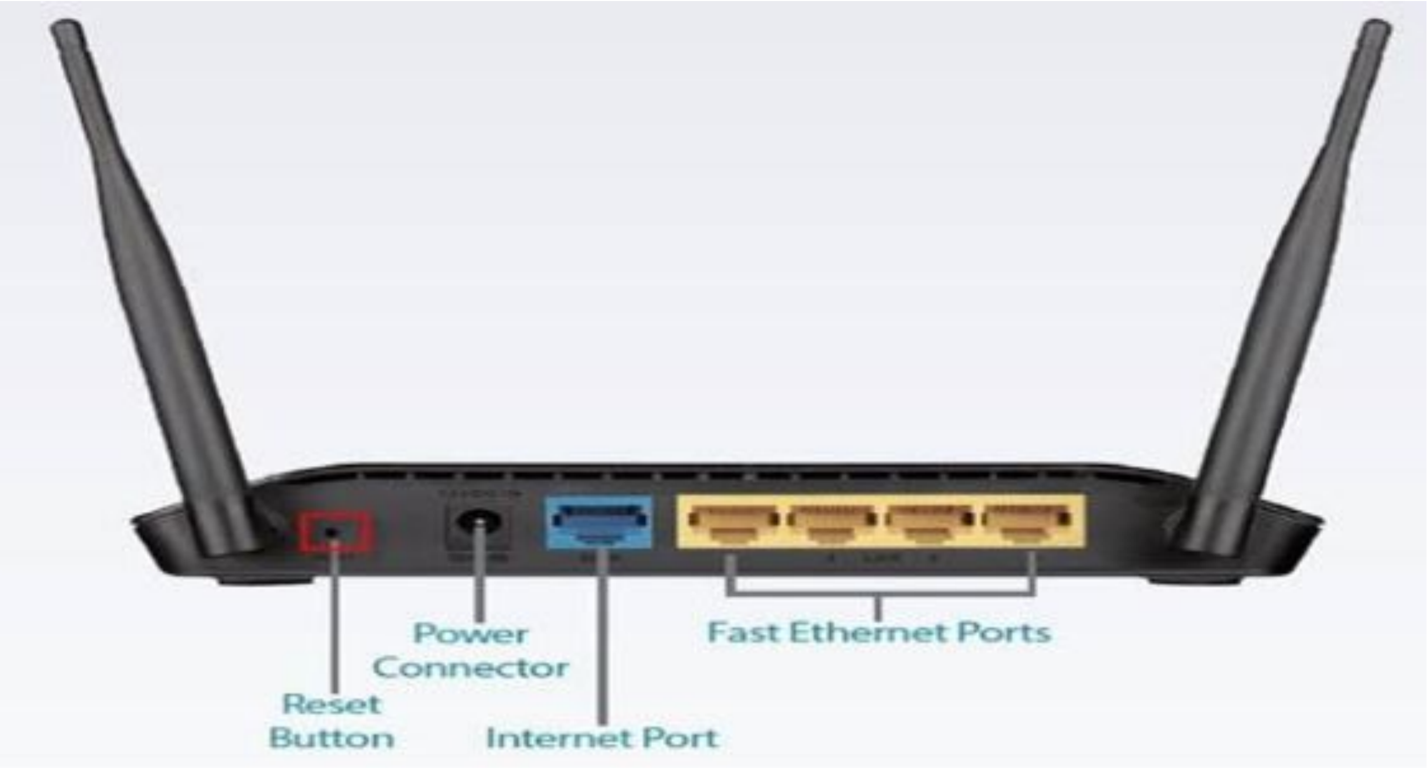
Router:

- A router is a network device similar to a switch that **routes data packets** based on their **IP addresses .**
- The router is primarily a **Network Layer device.**
- A router is also known as an **intelligent device** because it can automatically calculate the **best route** to pass network packets from source to destination.
- A router examines a data packet's **destination IP address** and uses headers and forwarding tables to determine the best way to transfer the packets.
- It communicates between **two or more** networks using protocols such as **ICMP**.
- Router is used to create larger **complex networks** by complex traffic routing.

Router:


Host A
Router
Host B
192.168.0.0/24
10.0.0.0/24
192.168.0.15
10.0.0.20
172.16.0.0/24
Host C
172.16.0.50

Router:


Power
Connector
Fast Ethernet Ports
Reset
Button
Internet Port

Functionality of Router:

- When a router **receives** the data, it **determines** the destination address by reading the header of the packet.
- Once the address is determined, it searches in its **routing table** to get know how to reach the destination and then forwards the packet to the next hop on the route.
- Routing tables play a very pivotal role in letting the router making a decision.
- Thus a routing table is ought to be **updated** and complete.

Types of Routing:

- **Static Routing:**
  - In static routing, the routing information is fed into the routing tables **manually**.
  - It does not only become a **time-taking task** but gets prone to errors as well.
  - Thus static routing is feasible for **tiniest environments** with minimum of one or two routers.

- **Dynamic Routing:**
  - For **larger environment** dynamic routing proves to be the practical solution.
  - The purpose of these protocols is to enable the other routers to transfer information **about to other routers**, so that the other routers can build their own **routing tables**.

Router:

- **Advantages of Router:**
  - It offers **advanced routing, flow control, and traffic isolation.**
  - They are **configurable**, allowing the network administrator to create policies based on routing decisions.
  - It can select the best path across the internetwork using **dynamic routing algorithms**.
  - It can reduce network traffic by establishing **collision domains** as well as **broadcast domains.**

- **Disadvantages of Router:**
  - A router is more **expensive** than a bridge or a repeater.
  - They are **protocol-dependent** devices that must comprehend the protocol they are transmitting.
  - The router is **slower** than bridges or repeaters because it must analyze data transmission from the physical to the network layer.

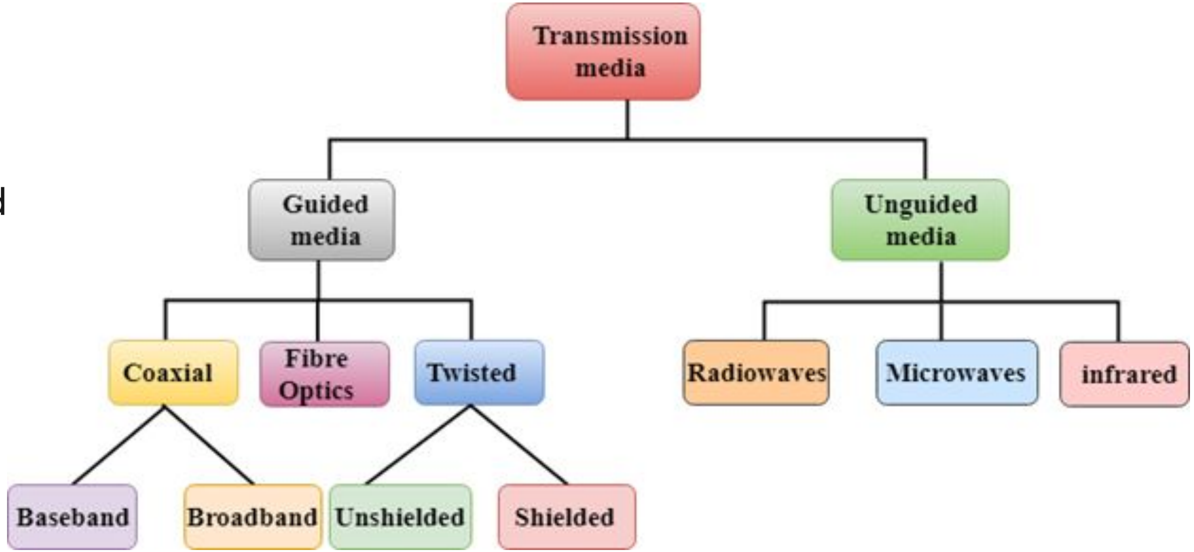
Transmission Media

2.2

▶ Transmission Media:

- ❖ Transmission media is the medium through which we send our data from one place to another i.e. one node to another node.
- ❖ Transmission media is the "connecting cables" or "connecting media" that connect two or more workstations.

- ❖ 2 Types:
  - ➢ Guided Media
  - ➢ Unguided Med



Guided Media(Bounded Media):

It is the transmission media in which signals are confined to a specific path from source to destination using wire or cable i.e. transmission is done via wire and Physical cable.

There are two types of guided media

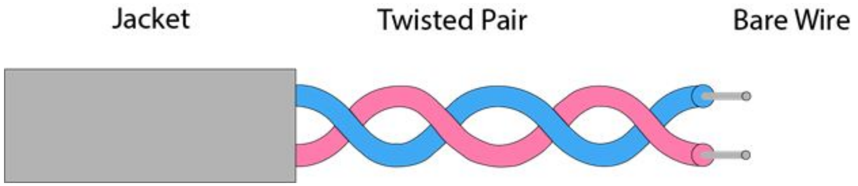
1. Twisted Pair Cable
    - Unshielded Twisted Pair(UTP)
    - Shielded Twisted Pair(STP)
2. Fiber Optic Cable
3. Coaxial Cable
    - Baseband Coaxial Cable
    - Broadband Coaxial Cable

Both UTP and STP are widely used in transmitting information across distance. UTP is a cable with wires that are twisted together to reduce noise generated by an external source.

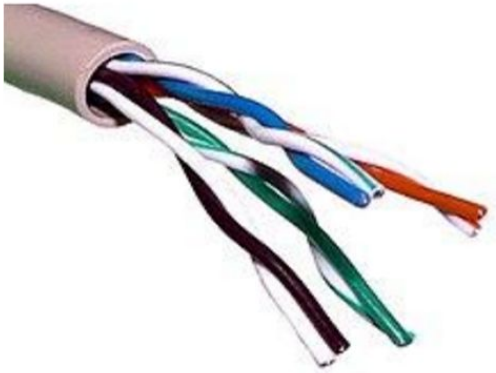
Twisted Pair Cable:

- Twisted pair is a physical media made up of a pair of cables twisted with each other.
- It is the most common form of wiring in data communication application.
- The wires come in pairs, The pairs of wires are twisted around each other.
- The degree of reduction in noise interference is determined by the number of turns per foot. Increasing the number of turns per foot decreases noise interference.
- The wires are copper wires. Copper wires are the most common wires used for transmitting signals because of good performance at low costs.

- Twisted Pair is of two types :
  - Unshielded Twisted Pair (UTP)
  - Shielded Twisted Pair (STP)



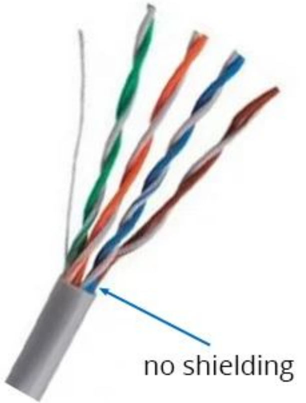
Twisted Pair Cable:



Some important points:

- Its frequency range is 0 to 3.5 kHz.
- Typical attenuation is 0.2 dB/Km @ 1kHz.
- Typical delay is 50 µs/km.
- Repeater spacing is 2km.

Twisted Pair Types:



Unshielded Twisted Pair:

- ❖ There is no special shield around the twisted wire pairs.
- ❖ Hence, it is called Unshielded Twisted Pair.
- ❖ It is more victim from EMI.

Shielded Twisted Pair:

- ❖ There is a braided shield around the twisted wire pairs which is also ground during installation.So,it is called Shielded Twisted Pair.
  - ■ Each twisted pair is shielded.
  - ■ Set of multiple twisted pairs in the cable is shielded.
  - ■ Each twisted pair and then all the pairs are shielded.
- ❖ Minimizes EMI radiation.

Twisted Pair Cables:

Parameter	UTP	STP
Full Form	Unshielded Twisted Pair	Shielded Twisted Pair
Structure	Cable with wires that are twisted together.	Cable(Wires) are enclosed in foil/shield.
Cost	Cheaper than STP	Costlier than UTP
Weight	Lighter than STP	Heavier than UTP
Noise and Interference	Prone to noise and interference.	Less prone to noise and interference.
Data Speed	Supports slower speed than on STP	Support higher speed than UTP
Grounding of Cable	Not required	Required
Target Deployments	Locations less prone to interference like offices and home.	Location prone to interference like factories and airports.
Categories	UTP cables categories as specified by EIA- Category-1, Category-2, Category-3, Category-4, Category-5, Category-5e, Category-6, Category-6a and Category-7.	Shielded cables have commonly these configurations- Foil Shielded and Braid Shielded.
Installation and Distance	Easy to install and distance travel is less.	Difficult to install and distance travel is more.

Straight through vs Crossover Wire:

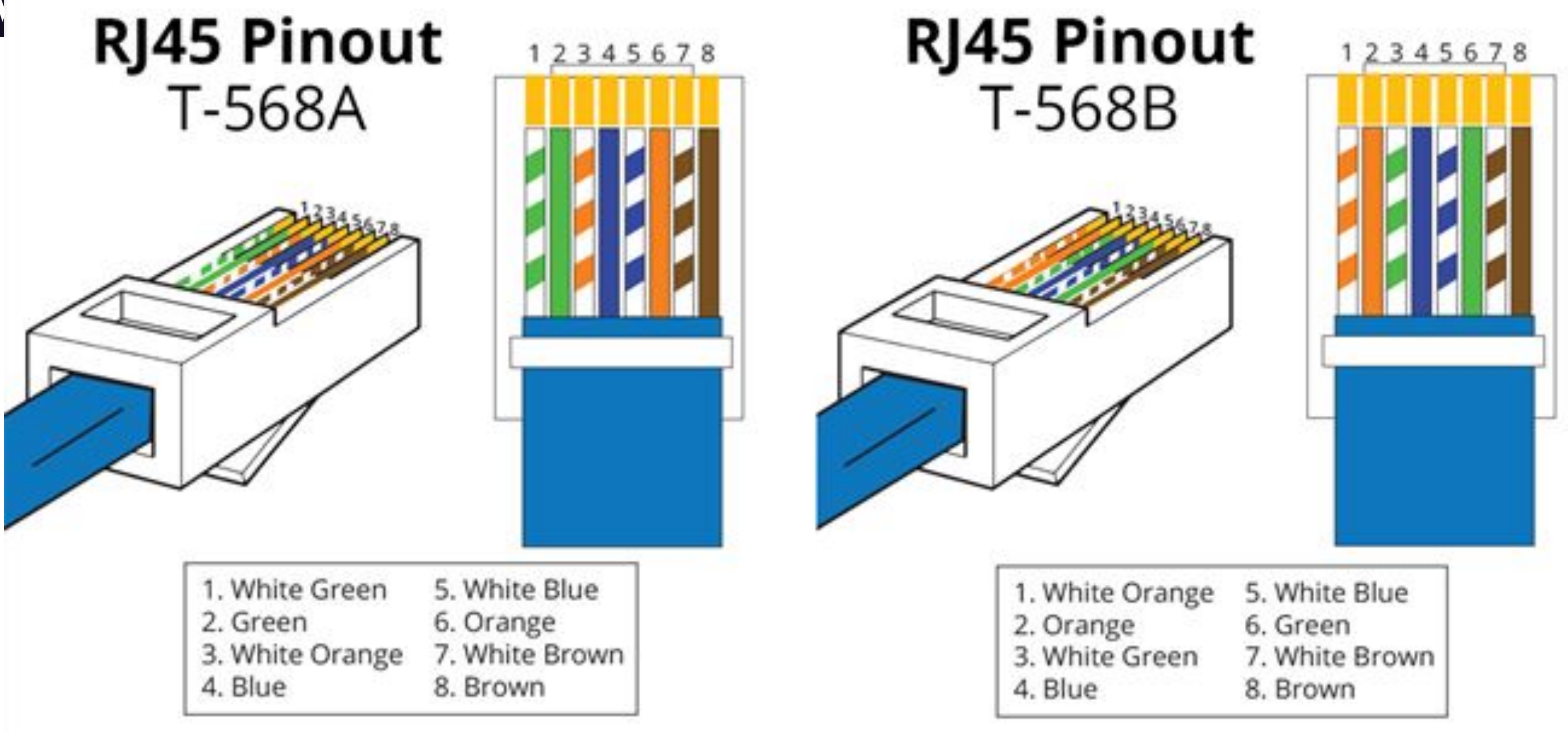
Straight-through Wire:

- ❖ Straight through cables are primarily used for connecting unlike devices.
- ❖ Use straight through Ethernet cable for the following cabling:
    - Switch to router
    - Switch to PC or server
    - Hub to PC or server

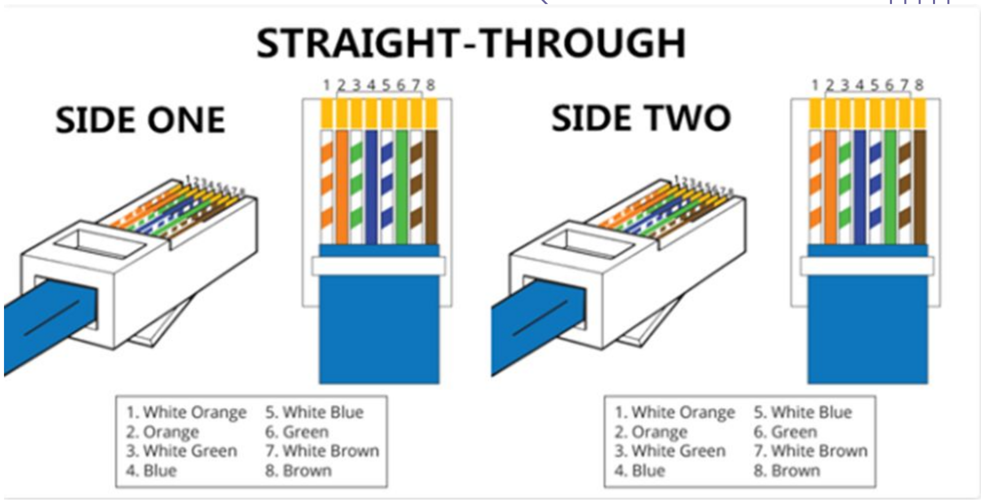
Cross-over Wire:

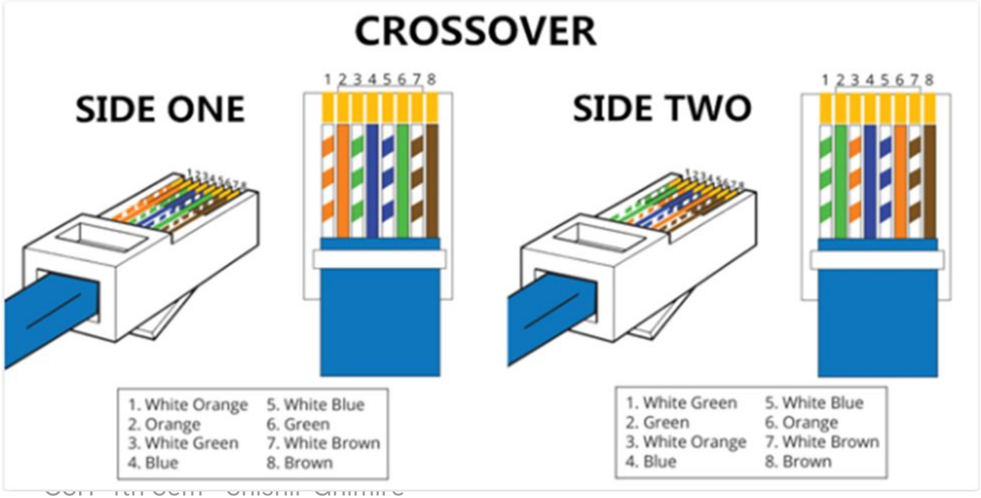
- ❖ Crossover cables are used for connecting alike devices.
- ❖ Use crossover wire for the following cabling:
    - Switch to switch
    - Hub to hub
    - Router to router
    - Router Ethernet port to PC NIC
    - PC to PC

Straight Through vs Crossover W



Straight Through vs Crossover Wire:





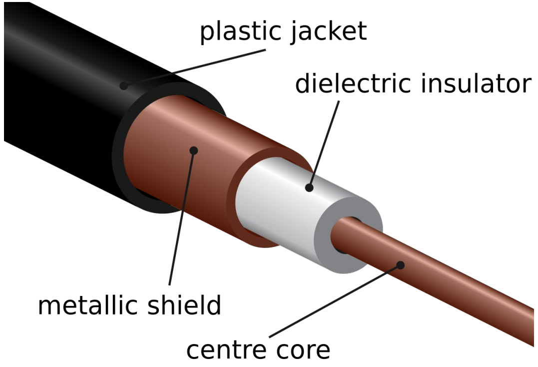
Coaxial Cable:

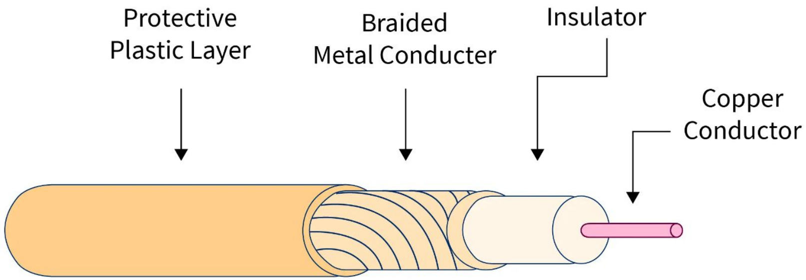
- Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- The name of the cable is coaxial as it contains two conductors parallel to each other.
- Coaxial cable consists of a solid wire core surrounded by one or more foil or wire shields, each separated by some kind plastic insulator.
- The inner core carries the signal and the shield provides the ground.
- The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- The coaxial cable has high electrical properties and is suitable for high speed communication.

Coaxial Cable:

Here the most common coaxial standards.

- ❖ RG(Radio Guide)
- ❖ **50-Ohm RG-7 or RG-11** : used with thick Ethernet.
- ❖ **50-Ohm RG-58** : used with thin Ethernet
- ❖ **75-Ohm RG-59** : used with cable television
- ❖ **93-Ohm RG-62** : used with ARCNET(Attached Resource Computer NETwork)





Coaxial Cable:

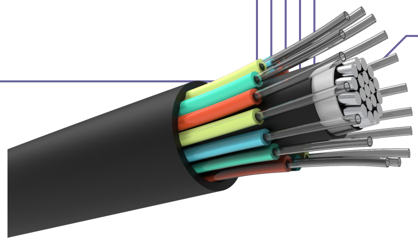
- **Baseband:**
  - Baseband is a communication techniques in which **digital signals** are placed in the transmission line **without change** in modulation.
  - It means that digital signal are directly transmitted over a transmission channel.
  - It transmits **only one** signals at a time.
  - Digital signals are also called Baseband Signals.

- **Broadband:**
  - A multiple data channel system in which the bandwidth of the transmission medium carries **several data** streams at the same time.
  - Data may be voice and video.
  - It can send data **by modulating** each signal onto a different frequency.

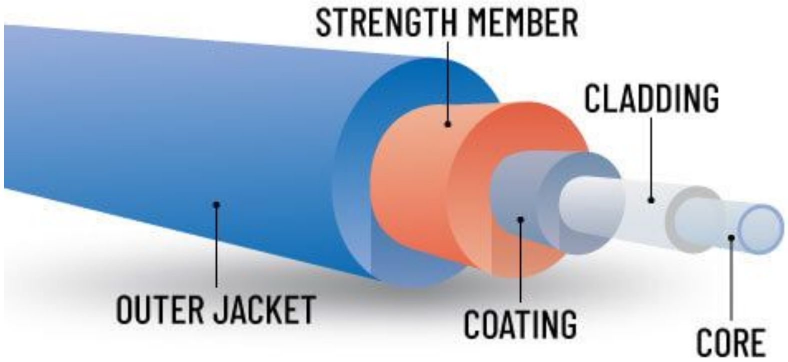
Coaxial Cable:

- **Advantages:**
  - Used in long distance telephone lines.
  - The coaxial cables offer high bandwidth as compared to other data transmission cables such as twisted pair cables.
  - Much higher noise immunity
  - Data transmission without distortion.
  - The coaxial cables can be used to deal with both analog signals as well as digital signals.
- **Disadvantages:**
  - Single cable failure can fail the entire network.
  - Difficult to install and expensive when compared with twisted pair.
  - If the shield is imperfect, it can lead to grounded loop.

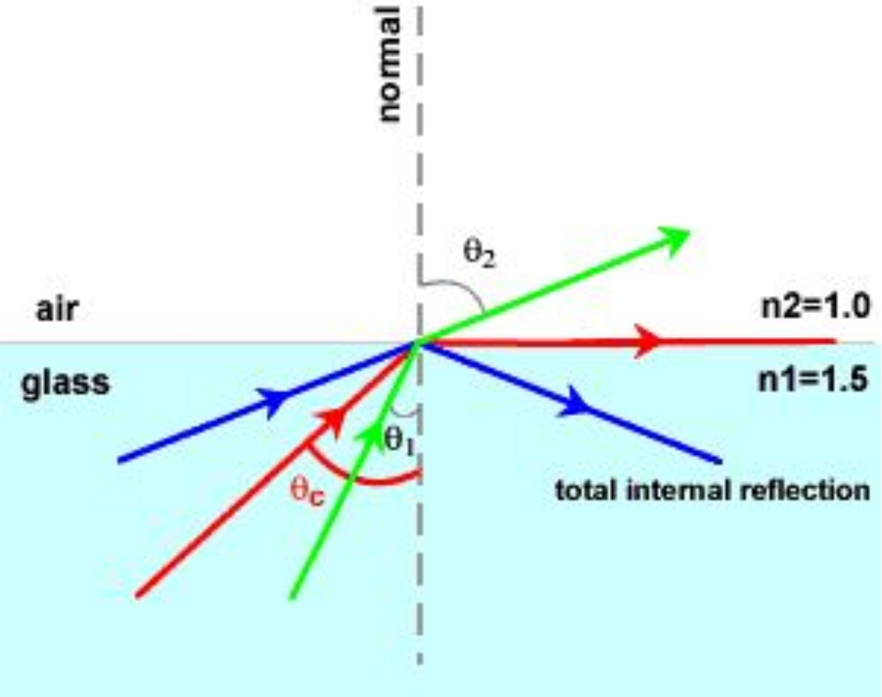
▶ Fiber Optic Cable:

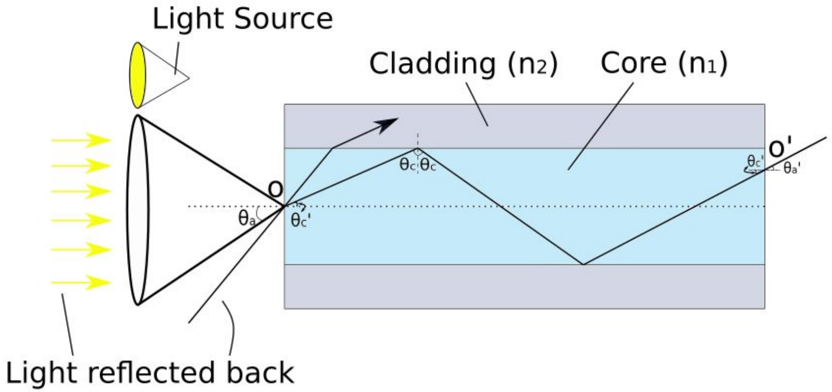


- ❖ Fibre optic cable is a cable that uses **optical signals** for communication.
- ❖ A fiber optic cable is made of high quality of the **glass or plastic** and is used to transfer digital data signals in the form of **pulses of light** up to distance of thousand of miles.
- ❖ Fibres optic cable carry communication signals using pulses of light generated by small laser or light-emitting diodes(**LEDs**).
- ❖ The plastic coating protects the optical fibres from heat, cold, electromagnetic interference from other types of wiring.
- ❖ Fibre optics provide faster data transmission than copper wires.
- ❖ Fiber optic cable has bandwidth more than **2 gbps** (Gigabytes per Second)



▶ Fiber Optic Cable & TIR Mechanism:





Elements of Fiber Optic Cable:

1. **Core:** The optical fibre consists of a narrow strand of glass or plastic known as a core. A core is a light transmission area of the fibre. The more the area of the core, the more light will be transmitted into the fibre.

2. **Cladding:** The concentric layer of glass is known as cladding. The main functionality of the cladding is to provide the lower refractive index at the core interface as to cause the reflection within the core so that the light waves are transmitted through the fibre.

3. **Jacket:** The protective coating consisting of plastic is known as a jacket. The main purpose of a jacket is to preserve the fibre strength, absorb shock and extra fibre protection.

Optical Fiber:

Advantages of Optical Fiber :

- It is immune to electrical and magnetic interference.
- It is highly suitable for harsh industrial environments.
- Fibre optic cables can be used for broadband transmission where several channels are handled in parallel.
- It is also possible to mix data transmission channels with channels for telescope, graphics, TV and sound

Disadvantages of Optical Fiber :

- Difficult to install.
- Connecting either two fibres together on a light source to a fibre is a difficult process.
- Light can reach the receiver out of phase.
- Most expensive.

Unbounded / Unguided Media:

- Unguided media transport electromagnetic waves **without using a physical conductor.**
- This type of communication is often referred to as wireless communication.
- In unguided media, ***air*** is the media through which the electromagnetic energy can flow easily. Thus, is available to anyone who has the device capable of receiving it.
- **Three ways** for wireless data propagation
  - Radio wave
  - Microwave
  - Infrared

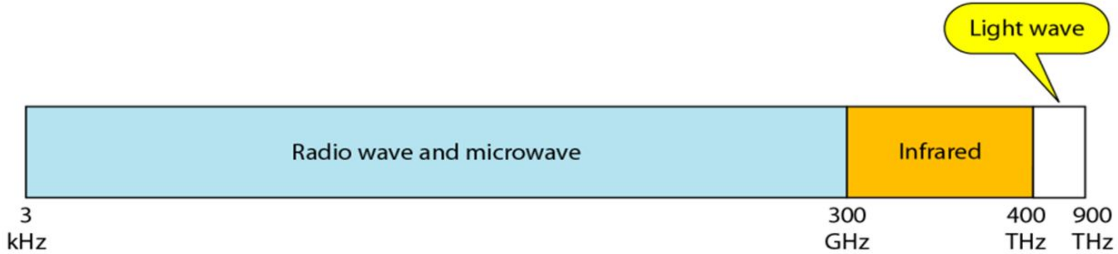


Figure : Electromagnetic spectrum for wireless communication

Infrared :

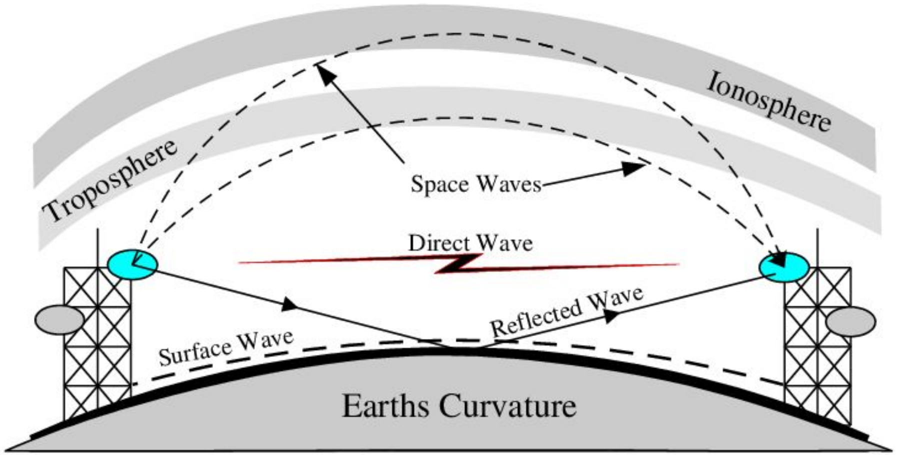
- An infrared transmission is a wireless technology used for communication over **short ranges**.
- The frequency of the infrared in the range from ***300 GHz to 400 THz***.
- It is used for short-range communication such as data transfer between two cell phones, TV remote operation, data transfer between a computer and cell phone resides in the same closed area.

Characteristics of Infrared:

- It supports **high bandwidth**, and hence the data rate will be very high.
- Infrared waves **cannot penetrate the walls.** Therefore, the infrared communication in one room cannot be interrupted by the nearby rooms.
- An infrared communication provides better security with minimum interference.
- Infrared communication is unreliable outside the building because the **sun rays** will interfere with the infrared waves.

Radiowave:

- Radio waves are the electromagnetic waves that are transmitted **in all the directions** of free space.
- Radio waves are **omnidirectional**, i.e., the signals are propagated in all the directions.
- The range in frequencies of radio waves is from **3 Khz to 1 Ghz**.
- In the case of radio waves, the **sending and receiving antenna are not aligned**, i.e., the wave sent by the sending antenna can be received by any receiving antenna.
- An example of the radio wave is FM radio.



Radiowave:

- **Applications of Radio Waves:**
  - A Radio wave is useful for **multicasting** when there is one sender and many receivers.
  - An FM radio, television, cordless phones are examples of a radio wave.

- **Advantages of Radio Waves:**
  - Radio transmission is mainly used for wide area networks and mobile cellular phones.
  - Radio waves **cover a large area**, and they can penetrate the walls.
  - Radio transmission provides a higher transmission rate.

- **Disadvantages of Radio Waves:**
  - The range of frequencies that can be accessed by existing technology is limited, so there is competition amongst companies for the use of frequencies.
  - Travel in a straight line, so repeater stations might be needed.

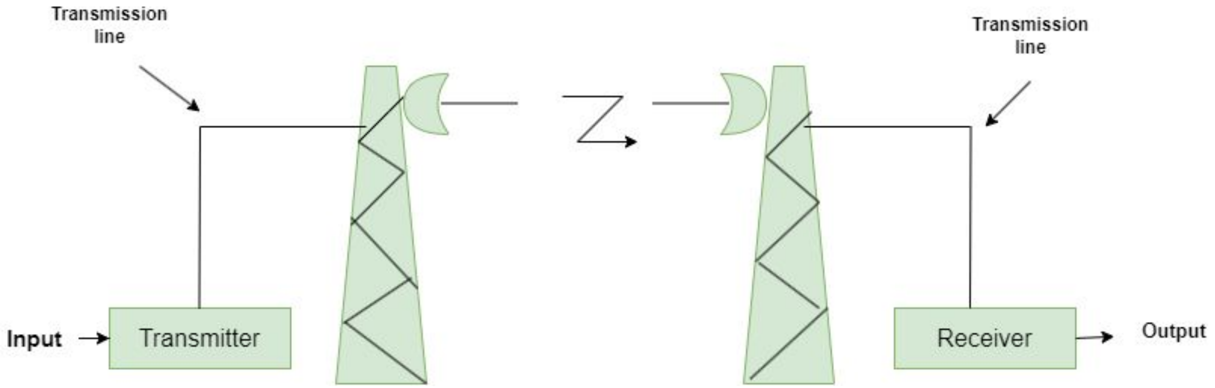
Microwave:

- Microwaves are the electromagnetic waves having the frequency in the range from **1GHz to 300 GHz.**
- Microwaves are **unidirectional** as the sending and receiving antenna is to be aligned, i.e., the waves sent by the sending antenna are narrowly focussed.
- Microwave signals are similar to radio and television signals and are used for long distance communication.
- Microwave transmission consists of ***transmitter, receiver and the atmosphere.***
- Microwaves are used for **unicast communication** such as cellular telephones, satellite networks, and wireless LANs.
- Higher frequency ranges **cannot** penetrate walls.
- Use **directional antennas** - point to point line of sight communications.
- It works on the **line of sight transmission**, i.e., the antennas mounted on the towers are the direct sight of each other.

Microwave:

Characteristics:

- **Frequency Range:** The frequency range of terrestrial microwave is from 4-6 GHz to 21-23 GHz.
- **Bandwidth:** It supports the bandwidth from 1 to 10 Mbps.
- **Short distance:** It is inexpensive for short distance.
- **Long distance:** It is expensive as it requires a higher tower for a longer distance.
- **Attenuation:** Attenuation means loss of signal. It is affected by environmental conditions and antenna size.



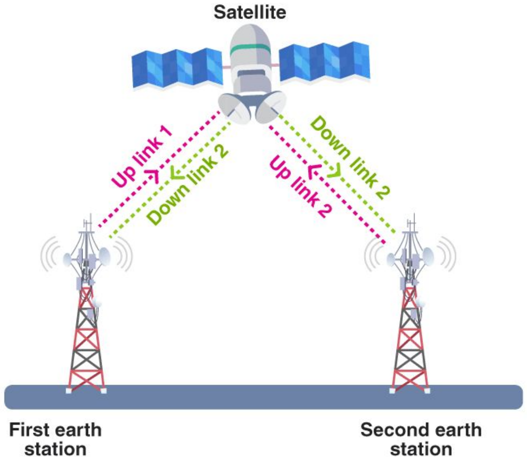
Microwave:

- **Advantages Of Microwave:**
  - Microwave transmission is cheaper than using cables.
  - It is free from land acquisition as it does not require any land for the installation of cables.
  - Microwave transmission provides an easy communication in terrains as the installation of cable in terrain is quite a difficult task.
  - Communication over oceans can be achieved by using microwave transmission.

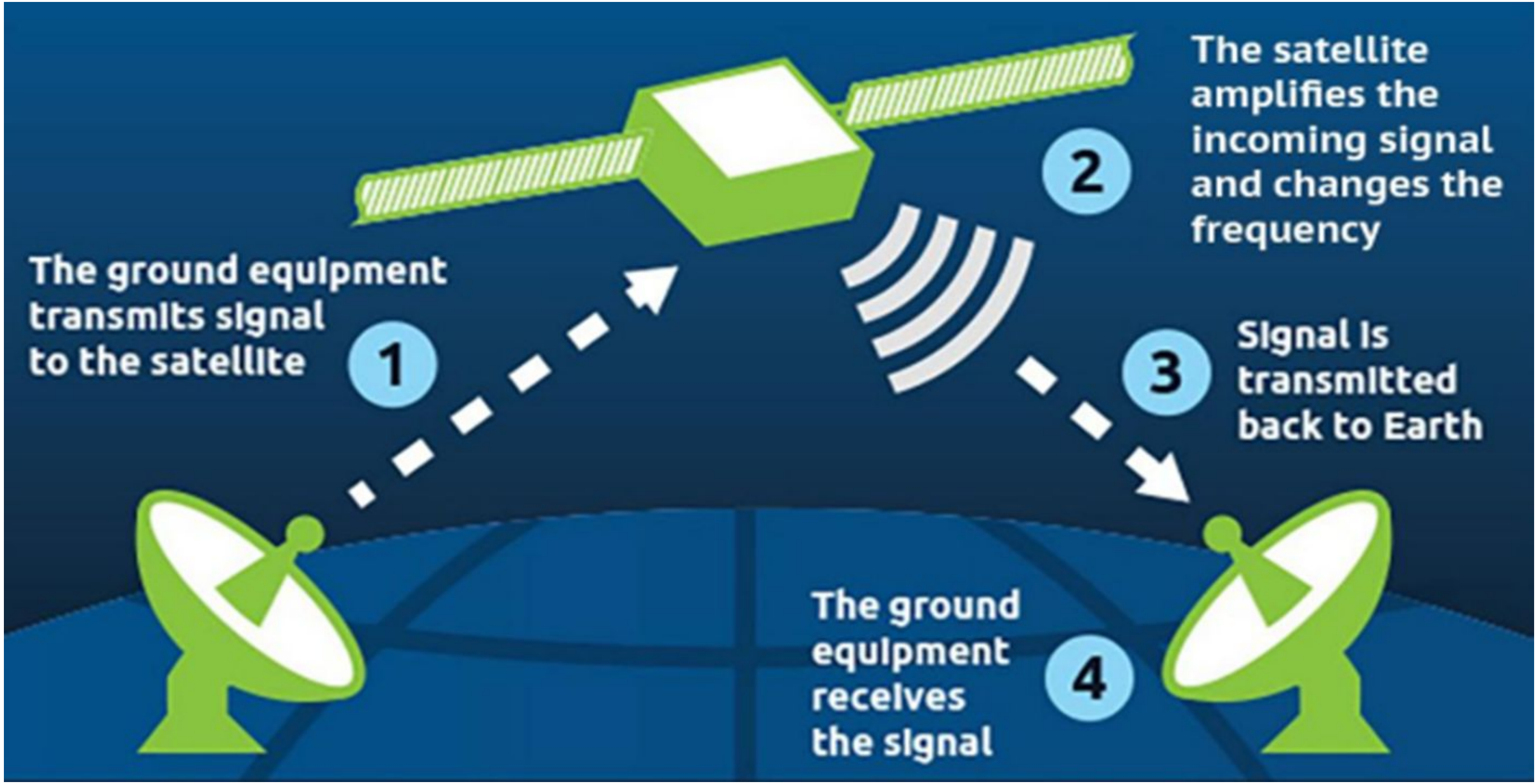
- **Disadvantages of Microwave:**
  - **Eavesdropping:** An eavesdropping creates insecure communication. Any malicious user can catch the signal in the air by using its own antenna.
  - **Out of phase signal:** A signal can be moved out of phase by using microwave transmission.
  - **Susceptible to weather condition:** A microwave transmission is susceptible to weather condition. This means that any environmental change such as rain, wind can distort the signal.
  - **Bandwidth limited:** Allocation of bandwidth is limited in the case of microwave transmission.

Satellite Communication:

- A communications satellite is an **artificial satellite** that relays and amplifies radio telecommunications signals via a transponder.
- It creates a communication channel between a source transmitter and a receiver at different locations on Earth.
- Communications satellites are used for television, telephone, radio, internet, and military applications.



Satellite Communication:



Satellite Communication:

Advantages of satellite communication:

- The area coverage through satellite communication is quite large.
- The satellite proves to be best alternative where the laying and maintenance of intercontinental cable is difficult and expensive.
- The heavy use of intercontinental traffic makes the satellite communication attractive.
- Satellite can cover large areas of the earth.

Disadvantages of satellite communication:

- Technological limitations prevent the deployment of large, high gain antennas on the satellite platform.
- Over crowding of available bandwidths due to low antenna gains.
- High investment cost and insecure cost associated with significant probability of failure.

Satellite Types:

Parameter	LEO	MEO	GEO
Satellite Height	500-1500 km	5000-12000 km	35,800 km
Orbital Period	10-40 min.	2-8 hours	24 hours
Number of Satellites	40-80	8-20	3
Satellite Life	Short	Long	Long
Number of Handoffs	High	Low	Least(none)
Gateway Cost	Very expensive	Expensive	Cheap
Propagation Loss	Least	High	Highest

LEO - Low Earth Orbit
MEO- Medium Earth Orbit
GEO- Geostationary Orbit

Ethernet Cable Standards

2.3

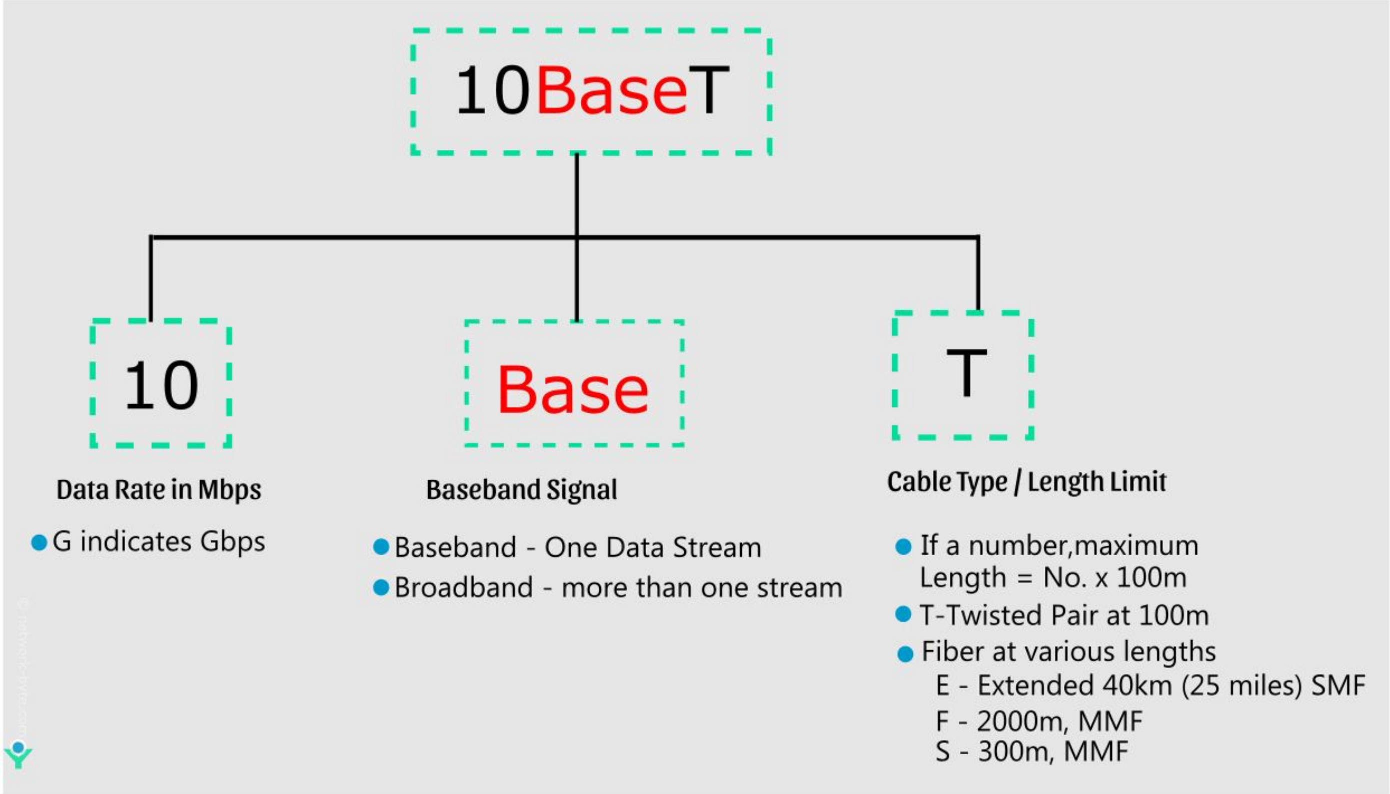
Overview

- Ethernet, developed by the Electrical and Electronic Engineers Institute, IEEE Standard 802, is the most popular LAN (local area network) technology used today.
- It defined the number of conductors that are required for a connection, the performance thresholds that can be expected, and provides the framework for data transmission.
- Several Standards are available in real market some of them are listed below.
    - UTP Standards
    - Fiber Optic Standards

Ethernet Standards:

Ethernet Type	Bandwidth	Cable Type	Maximum Distance
10Base-T	10Mbps	Cat 3/Cat 5 UTP	100m
100Base-TX	100Mbps	Cat 5 UTP	100m
100Base-TX	200Mbps	Cat 5 UTP	100m
100Base-FX	100Mbps	Multi-mode fiber	400m
100Base-FX	200Mbps	Multi-mode fiber	2Km
1000Base-T	1Gbps	Cat 5e UTP	100m
1000Base-TX	1Gbps	Cat 6 UTP	100m
1000Base-SX	1Gbps	Multi-mode fiber	550m
1000Base-LX	1Gbps	Single-mode fiber	2Km
10GBase-T	10Gbps	Cat 6a/Cat 7 UTP	100m
10GBase-LX	10Gbps	Multi-mode fiber	100m
10GBase-LX	10Gbp	Single-mode fiber	10Km

Ethernet Standards > Nomenclature:



UTP Standards:

The following are a few UTP Cable standards used in the market:

- **Cat 3 UTP**
  - Category 3 UTP is rated to carry data up to 10Mbit/s.
  - Cat 3 UTP was the standard cable used with Ethernet 10Base-T.
  - Category 3 UTP is an early standard, suitable for basic networking applications.
- **Cat 5 UTP**
  - Category 5 UTP is rated to carry Ethernet up to 100Mbit/s and ATM up to 155Mbit/s.
  - Cat 5 UTP was the standard cable used with Ethernet 100Base-TX.
  - Category 5 UTP improved upon Cat 3 and became a widely adopted standard for Fast Ethernet networks.
- **Cat 5e UTP**
  - Category 5e UTP is an enhanced version of Cat 5 UTP.
  - Cat 5e UTP is rated to carry data up to 1000Mbit/s.
  - Cat 5e UTP is the standard cable used with Ethernet 1000Base-T.

UTP Standards:

- **Cat 5e UTP**
  - Category 5e UTP addresses crosstalk and interference issues, providing improved performance over Cat 5.
  - It became the standard for Gigabit Ethernet.

- CAT 6
  - Very similar to Cat 5 UTP but designed and manufactured to higher standards.
  - Capable of supporting higher data rates than Cat 5.

UTP Standards:

UTP Categories - Copper Cable				
UTP Category	**Data Rate**	**Max. Length**	**Cable Type**	**Application**
CAT1	Up to 1Mbps	-	Twisted Pair	Old Telephone Cable
CAT2	Up to 4Mbps	-	Twisted Pair	Token Ring Networks
CAT3	Up to 10Mbps	100m	Twisted Pair	Token Rink & 10BASE-T Ethernet
CAT4	Up to 16Mbps	100m	Twisted Pair	Token Ring Networks
CAT5	Up to 100Mbps	100m	Twisted Pair	Ethernet, FastEthernet, Token Ring
CAT5e	Up to 1 Gbps	100m	Twisted Pair	Ethernet, FastEthernet, Gigabit Ethernet
CAT6	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (55 meters)
CAT6a	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (55 meters)
CAT7	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (100 meters)

UTP Termination:

- **Two-Pair (Four-Wire) UTP (Telephone Use):**
  - Connector: Normally terminated in an **RJ-11** connector.
  - Usage: Commonly used for telephone applications.

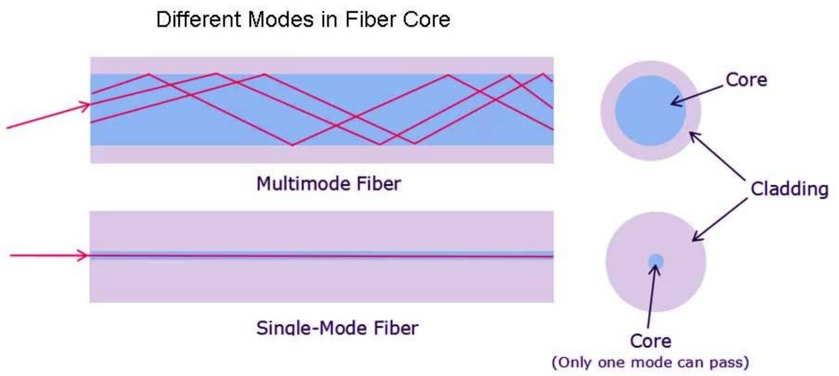
- **Four-Pair (Eight-Wire) UTP:**
  - Connector: Normally terminated in an **RJ-45** connector.
  - Usage: Commonly used for data applications, including networking.

Fiber Optic Standards:

- This cable consists of core, cladding, buffer, and jacket.
- The core is made from the thin strands of glass or plastic that can carry data over the long distance.
- The core is wrapped in the cladding; the cladding is wrapped in the buffer, and the buffer is wrapped in the jacket.
  - Core carries the data signals in the form of the light.
  - Cladding reflects light back to the core.
  - Buffer protects the light from leaking.
  - The jacket protects the cable from physical damage.
- Fiber optic is completely immune to (EMI) and Radio Frequency Interference (RFI).
- It reflects light from one endpoint to another based on how many beams of light are transmitted at a given time, there are two types of fiber optical cable;
  - **SMF:** (Single-Mode Fiber)
  - **MMF:**(Multi-Mode Fiber)

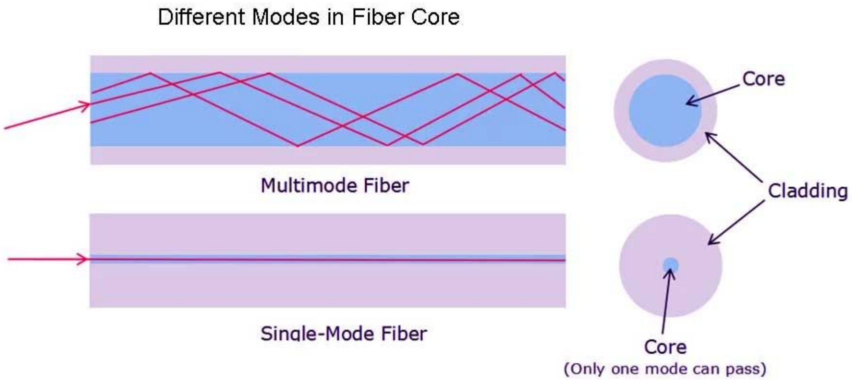
Single-Mode Fiber Optical Cable:

- ❖ The single-mode core is smaller and can carry light directly over long distances, making it suitable for applications requiring high-speed and low data loss.
- ❖ Single-mode cables are more expensive, and data transmission may be limited due to their size.
- ❖ Less data loss and Interference.
- ❖ This cable uses a laser as the light source and transmits 1300 or 1550 nm wavelengths of light.



Multi-Mode Fiber Optical Cable:

- ❖ This cable carries multiple beams of light.
- ❖ Due to multiple beams, it carries much more data than the SMF cable.
- ❖ Used in shorter distances.
- ❖ Typically uses LEDs (Light Emitting Diodes) or laser diodes as light sources.
- ❖ Uses an LED as the light source and transmits 850 or 1300 nm wavelengths of light.



Fiber Optic Standards:

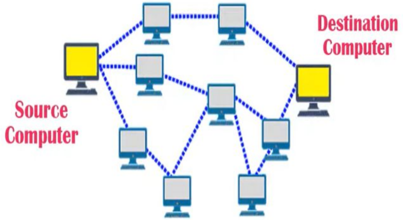
Cabling Standard	Cabling Type	Max Reach
10GBASE-SR	62.5µm OM3 multimode fiber	300m
	50µm OM4 multimode fiber	400m
10GBASE-LR	9µm single-mode fiber	10km
10GBASE-ER	9µm single-mode fiber	40km
10GBASE-ZR	9µm single-mode fiber	80km
10GBASE-LX4	9µm single-mode fiber	10km
	62.5µm multimode fiber	300m
	50µm multimode fiber	
10GBASE-LRM	9µm single-mode fiber	220m
10GBASE-T	Cat 6, Cat 6a or 7 twisted pair	30m
10G DAC/AOC	Copper RJ45	1-10m/up to 20m

Circuit, Message & Packet Switching

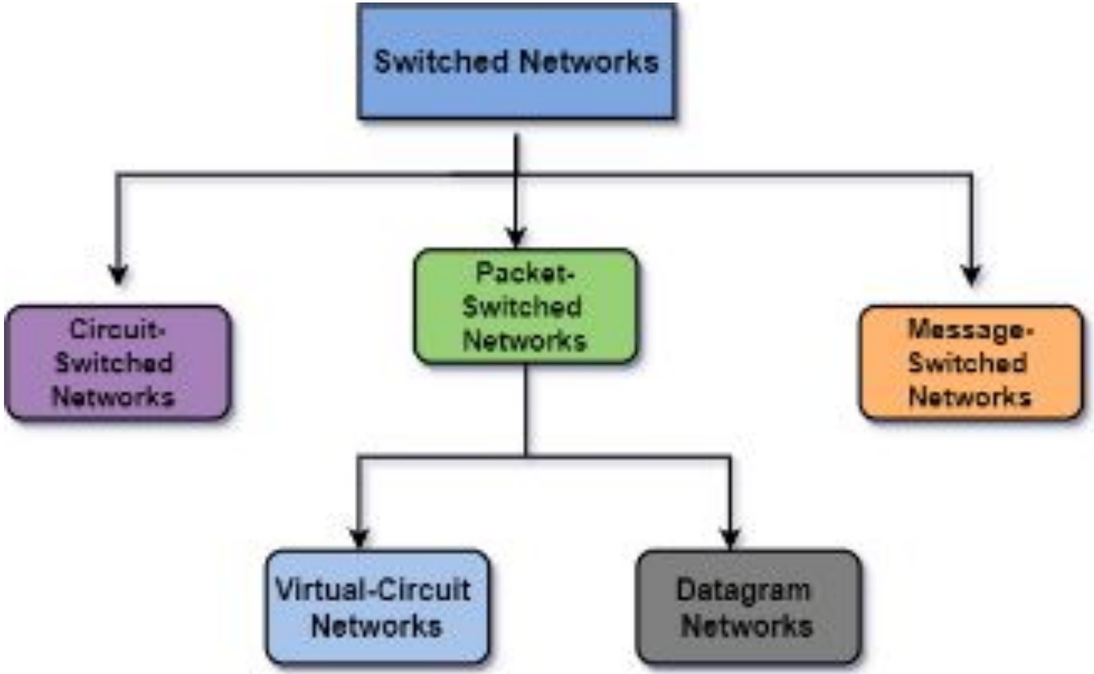
2.4

Switching:

- Switching is process to forward packets coming in from **one port to a port** leading towards the destination.
- Switching in a computer network is achieved by using switches.
- **Switch:** A switch is a small hardware device which is used to join multiple computers together with one local area network (LAN).
- Switching helps in deciding the best route for data transmission if there are multiple paths in a larger network.
- **For example:** whenever a telephone call is placed, there are numerous junctions in the communication path that perform this movement of data from one network onto another network."
- A switched network basically consists of a **series of interlinked nodes.** These interlinked nodes are known as switches.



Switched Network:



Switched Network:

Advantages:

- ❖ There is an increase in the available bandwidth on the network by using switches.
- ❖ There will be less frame collision as switch creates the collision domain for each connection.
- ❖ It reduces the workload on individual PCs as it sends the information to only that device which has been addressed.
- ❖ It increases the overall performance of the network by reducing the traffic on the network.

Disadvantages:

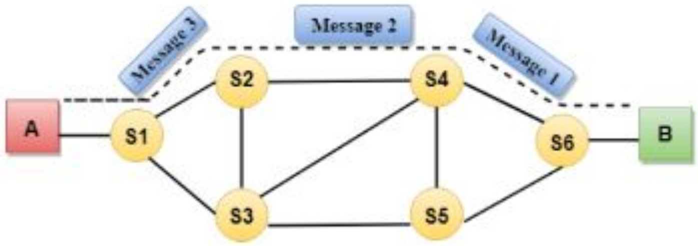
- ❖ A switch is more expensive than network bridges.
- ❖ It is difficult to trace the connectivity issues in the network through a switch.
- ❖ In order to handle multicast packets, proper design and configuration are needed.

Circuit Switching:

- Circuit-switching is a real-time connection-oriented system in which a dedicated channel (or circuit) is set up for a single connection between the sender and recipient during the communication session.
- There is a need of a pre-specified route from which data will travels and no other data is permitted.(Other devices can't use this route for data transmission because this route is reserved.)
- The Circuit Switching technique is mainly used in the public telephone network for voice communication as well as for data communication.
- **Example:** In the telephone communication system, the normal voice call is an example of Circuit Switching. The telephone service provider maintains an unbroken link for each telephone call.

Circuit Switching:

1. **Establish a circuit**
   - In this phase, there is an establishment of the circuit which simply means a dedicated link is established between the sender and the receiver with the help of several switching centers or nodes.
2. **Transfer the data**
   - Once the circuit is established, the connection is established which means that data transfer can take place between sender and receiver.
3. **Disconnect the circuit**
   - On the completion of communication between the sender and receiver, the circuit disconnects. In order to disconnect a signal is sent either by the sender or receiver



Circuit Switching:

Advantages:

- ❖ As a dedicated link between the sender and the receiver it provides a guarantee of dedicated transmission.
- ❖ There is a dedicated path between sender and receiver thus there are no chances for the delay.
- ❖ It is best for long transmission because it facilitates a dedicated link between sender and receiver.

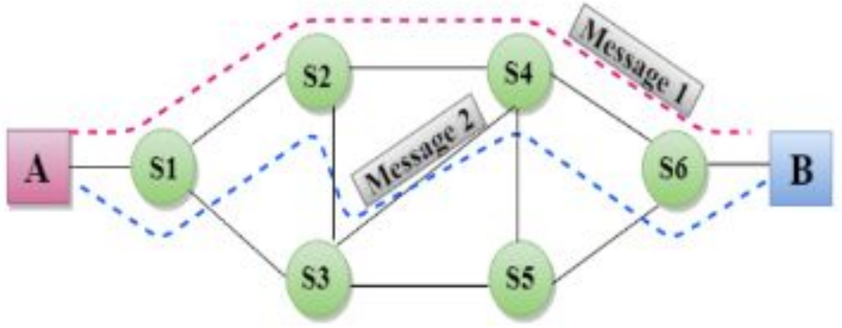
Disadvantages:

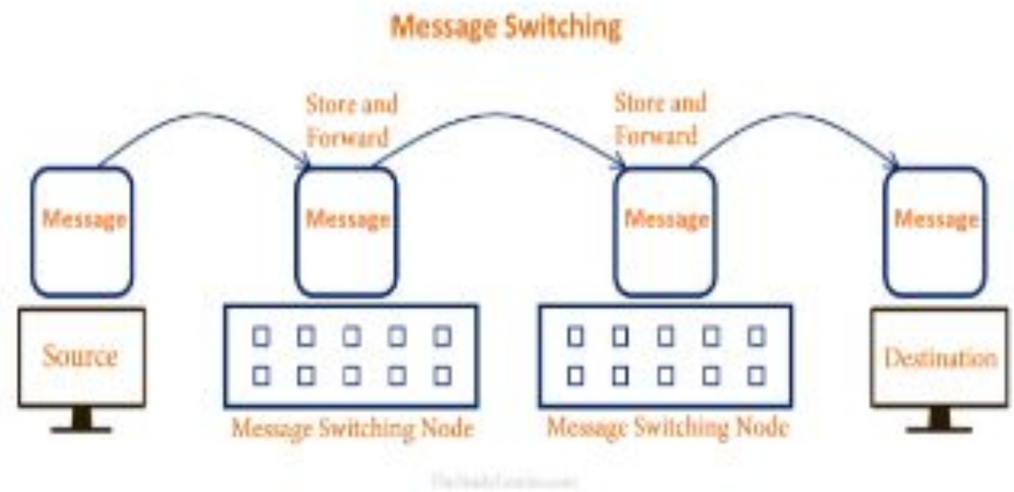
- ❖ There is a dedicated path between sender and receiver; thus this path is reserved for these two particular devices and cannot be used by any other device.
- ❖ It becomes inefficient if the connection is established between sender and receiver but there is no data transfer between them.
- ❖ As there is a dedicated path between sender and receiver; thus, this technique is expensive.

Message Switching:

- Message Switching is a switching technique in which a message is transferred as a complete unit and routed through intermediate nodes at which it is stored and forwarded.
- Message switching does not set up a dedicated channel (or circuit) between the sender and recipient during the communication session as the destination address is appended to the message.
- This technique was somewhere in middle of circuit switching and packet switching
- A message-switching node working on message-switching, first receives the whole message and stores it entirely on the disk and then transfers the whole message to the next message-switching node and so on until the message reaches the destination.
- If the next message-switching node does not have enough resources to accommodate large-size messages, the message is stored and the previous switching node has to wait. Hence it is called **Store and Forward Network**."

Message Switching:





Message Switching:

Advantages:

- Data channels are shared among the communicating devices that improve the efficiency of using available bandwidth.
- Traffic congestion can be reduced because the message is temporarily stored in the nodes.
- Message priority can be used to manage the network.
- The size of the message which is sent over the network can be varied. Therefore, it supports the data of unlimited size.

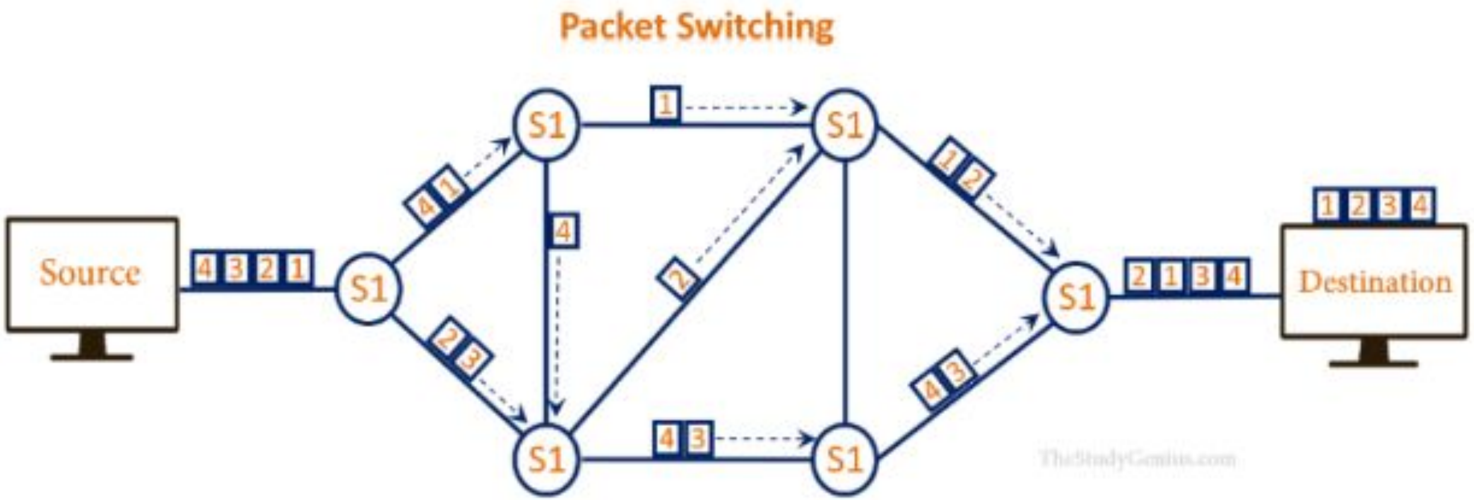
Disadvantages:

- The message switches must be equipped with sufficient storage to enable them to store the messages until the message is forwarded.
- The Long delay can occur due to the storing and forwarding facility provided by the message switching technique.
- The message-switched networks are very slow in nature because the processing takes place in each and every node and thus it may result in poor performance.

Packet Switching:

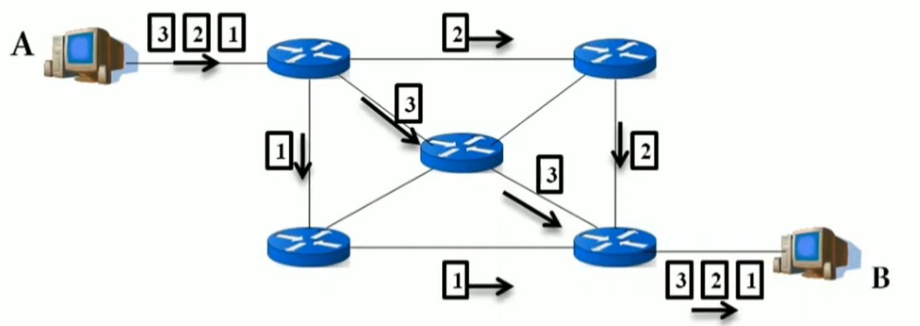
- Shortcomings of message switching gave birth to an idea of packet switching.
- Packet Switching is a technique of switching in which the message is divided into smaller pieces that are known as packets.
- Every packet contains a header that comprises of 2 things: Header and Payload.
  - **Header:** contains the routing information
  - **Payload:** contains the data that is to be transferred
- Every packet contains some information in its headers such as source address, destination address, and sequence number.
- Packets will travel across the network, taking the shortest path possible(multiple paths is possible that cannot be in order).
- All the packets are reassembled at the receiving end in the correct order. If any packet is missing or corrupted, then the message will be sent to resend the message.
- If the correct order of the packets is reached, then the acknowledgment message will be sent.

Packet Switching:



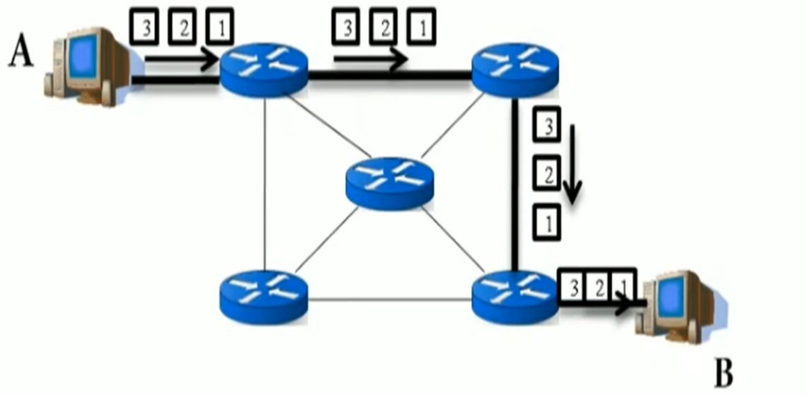
Datagram Packet Switching:

- Datagram Packet Switching is also known as connectionless switching which means the path is not fixed.
- Intermediate nodes take the routing decisions to forward the packets.
- It is a packet-switching technology in which a packet is known as a datagram, is considered as an independent entity.
- Each packet contains information about the destination and the switch uses this information to forward the packet to the correct destination.
- The packets are reassembled at the receiving end in correct order.



Virtual Circuit Switching:

- Virtual Circuit Switching is also known as connection-oriented switching.
- In the case of Virtual circuit switching, a preplanned route is established before the messages are sent.
- Call request and call accept packets are used to establish the connection between sender and receiver.
- In this case, the path is fixed for the duration of a logical connection.
- There are three phases in VCS:
  - Setup phase(Virtual Path created)
  - Data transfer phase
  - Teardown phase (teardown frame)



▶ Comparison:

Parameter	Datagram Switching	Virtual Circuit
Addressing	Each packet contains source and destination address with sequence number.	Each Packet contains Short VCI number
State information	Does not hold Packet level state information	Information about each VC is maintained
Routing	Each packet is routed independently.	Route is chosen when VCI is setup. All packets follows this route.
Congestion Control	Difficult	Easy if enough buffers can be allocated in advance.
Resource failure	Packet are lost only during resource failure.	All VCs passing through the failed resource are terminated.
Suitability	Generally Connectionless service	Connection oriented Service
Path	No fixed path	Fixed path
Packet receives	Out of order	Same order
Reliability	Less reliable	More reliable
Packet Loss	More	Less
Delay	More	Less
Resource	Resource are not reserved.	Resources are reserved.

Packet Switching:

Advantages:

- In packet switching technique, switching devices do not require massive secondary storage to store the packets, so cost is minimized to some extent.
- If any node is busy, then the packets can be rerouted which ensures reliable communication.
- It does not require any established path prior to the transmission, and many users can use the same communication channel simultaneously, hence makes use of available bandwidth very efficiently.

Disadvantages:

- Packet Switching technique cannot be implemented in those applications that require low delay and high-quality services.
- The protocols used in a packet switching technique are very complex and requires high implementation cost.
- If the network is overloaded or corrupted, then it requires retransmission of lost packets. It can also lead to the loss of critical information if errors are not recovered.
-

Comparison:

Basics	Circuit Switching	Message Switching	Packet Switching
Connection Creation	Connection is created between the source and destination by establishing a dedicated path between source and destination.	Links are created independently one by one between the nodes on the way.	Links are created independently one by one between the nodes on the way.
Queuing	No queue is formed.	Queue is formed.	Queue is formed.
Message and Packets	There is one big entire data stream called a message.	There is one big entire data stream called a message.	The big message is divided into a small number of packets.
Routing	One single dedicated path exists between the source and destination.	Messages follow the independent route to reach a destination.	Packets follow the independent path to hold the destination.
Addressing and sequencing	Messages need not be addressed as there is one dedicated path.	Messages are addressed as independent routes are established.	Packets are addressed, and sequencing is done as all the packets follow the independent route.
Propagation Delay	No	Yes	Yes
Transmission Capacity	Low	Maximum	Maximum
Sequence Order	Message arrives in Sequence.	Message arrives in Sequence.	Packets do not appear in sequence at the destination.
Use Bandwidth	Wastage	Bandwidth is used to its maximum extent.	Bandwidth is used to its maximum extent.

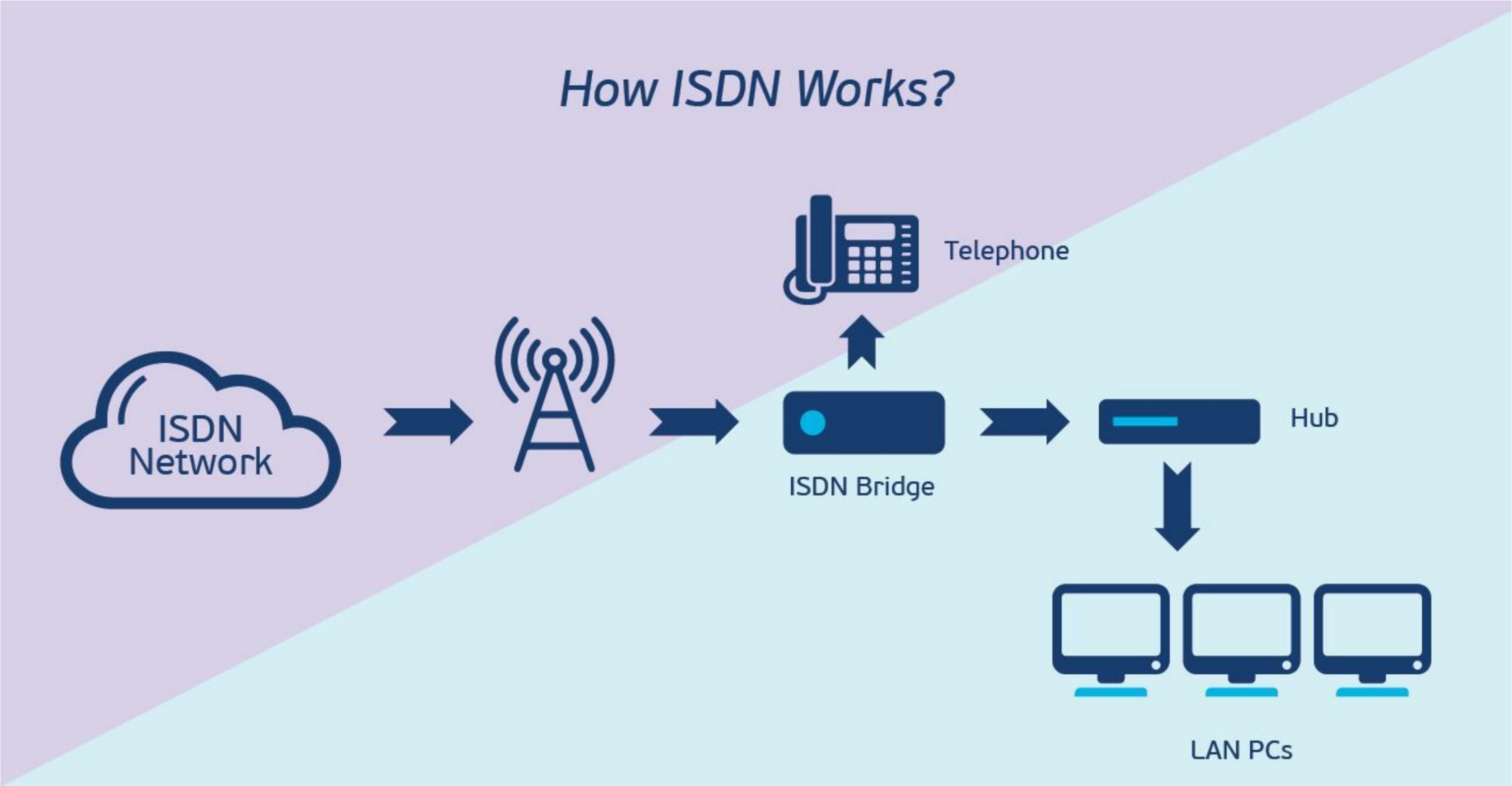
ISDN

2.5

ISDN (Integrated Services Digital Network):

- ISDN stands for Integrated Services Digital Network.
- These are a set of communication standards for simultaneous digital transmission of voice, video, data, and other network services over the traditional circuits of the public switched telephone network.
- Before Integrated Services Digital Network (ISDN), the telephone system was seen as a way to transmit voice, with some special services available for data.
- The main feature of ISDN is that it can integrate speech and data on the same lines, which were not available in the classic telephone system.
- The whole idea is to digitize the telephone network to permit the transmission of audio,video, image and text over existing telephone lines.
- ISDN is a circuit-switched telephone network system, but it also provides access to packet switched networks that allows digital transmission of voice and data.

ISDN (Integrated Services Digital Network):

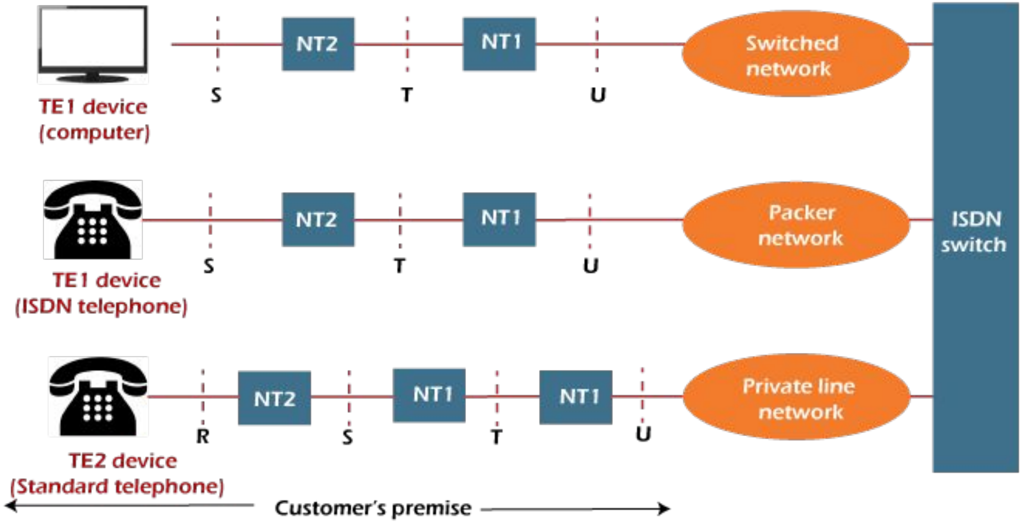


Principles of ISDN:

The various principles of ISDN as per ITU-T recommendation are:

- It supports both circuit switching & packet switching with the connections at 64 kbps.
- In ISDN layered protocol architecture is used for specification.
- ISDN services includes some network management functions.
- In ISDN network several configurations are possible for implementing.
- Support voice and non-voice applications

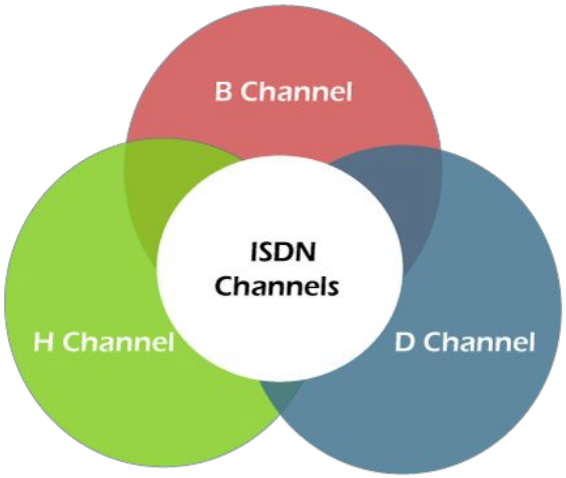
Principles of ISDN:



- R (Rate Transfer Point): Interface between non-ISDN devices and Terminal Adapter.
- S (System Transfer Point): Interface between user terminal and NT2, using a four-wire RJ-45 connector.
- T (Terminal Transfer Point): Interface between NT1 and NT2.
- U: Interface between NT1 and carrier network's line termination equipment.

ISDN CHANNELS::

- ISDN structure have a central ISDN office in which all the users are linked to this through a digital pipe.
- This digital pipe have different capacities and have a different data transfer rates and these are organized into multiple channels of different sizes.



ISDN CHANNELS::

1. B Channel: It stands for Bearer channel. It has a 64 kbps standard data rate. It is a basic user channel and can carry any digital information in full-duplex mode. In this transmission rate does not exceed 64 kbps. It can carry digital voice, digital data, and any other low data rate information.
2. D Channel: It stands for Data Channel. This channel carry control signal for bearer services. This channel is required for signaling or packet-switched data and all-controlling signals such as establishing calls, ringing, call interrupt, etc.
3. H Channel:It stands for Hybrid Channel. It provides user information at higher bit rates. There are 3 types of Hybrid Channel depending on the data rates. Following are the hybrid channels types:
    - Hybrid Channel 0 with 384 kbps data rate.
    - Hybrid Channel 11 with 1536 kbps data rate.
    - Hybrid Channel 12 with 1920 kbps data rate.

ISDN User Interfaces:

ISDN mainly offers following types of interface:

1. **Basic Rate Interface (BRI)**
2. **Primary Rate Interface (PRI)**

Base Rate Interface (BRI):

- BRI is a type of ISDN service designed for residential and small business use.It provides a lower level of service compared to PRI but is sufficient for most small-scale communication needs.
- It consists of two B channels (Bearer channels) for voice and data, each with a capacity of 64 kilobits per second (Kbps), and one D channel (Data channel) for signalling with a capacity of 16 Kbps.
- The total bandwidth of a BRI line is 144 Kbps (2B + 1D).
- Applications:
  - **Voice calls.**
  - **Internet access.**
  - **Fax transmission.**
  - **Video conferencing for small setups.**

Base Rate Interface (BRI):

Basic Rate Interface (BRI)

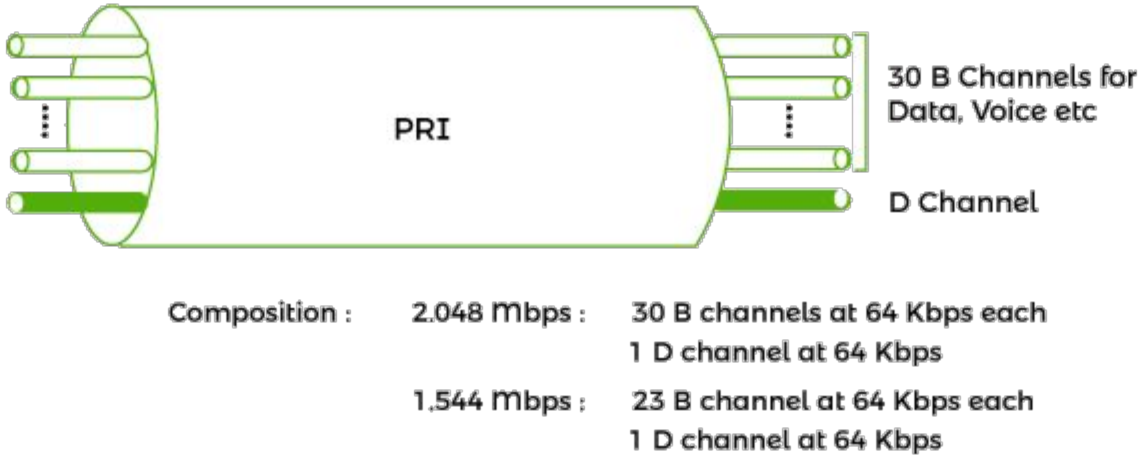
Advantages:

- Provides a reliable and relatively fast digital connection.
- Supports simultaneous voice and data transmission.
- Ideal for small-scale users needing integrated services.

Primary Rate Interface (PRI):

- PRI is a higher capacity ISDN service intended for larger organizations and businesses with higher communication needs.
- Structure:
  - **North America and Japan:** 23 B channels and 1 D channel (23B+D), using T1 lines.
  - **Europe and other regions:** 30 B channels and 1 D channel (30B+D), using E1 lines.
  - **B Channels:** Each B channel provides 64 Kbps for voice, video, data, and other services.
  - **D Channel:** The D channel, at 64 Kbps, handles signaling and control information.
- The total bandwidth of a PRI line is higher than BRI and depends on the number of B channels available (e.g., 23B + 1D in a North American PRI, resulting in a total bandwidth of 1.544 Mbps).

Primary Rate Interface (PRI):



Applications:

- Large-scale voice and data transmission.
- Video conferencing with multiple participants.
- High-speed internet access.

Advantages:

- Higher capacity and bandwidth, supporting more simultaneous connections.
- Reliable and robust for high-demand environments.
- Efficient management of multiple calls and data streams.

ISDN Services:

ISDN mainly offers following types of services:

1. **Bearer Services**
2. **Tele Services**
3. **Supplementary Services**

Bearer Services:

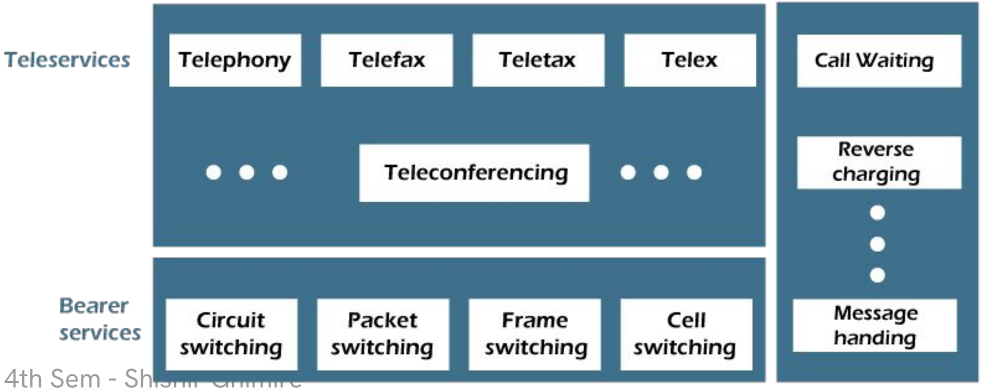
- Transfer of information (voice, data, and video) between users without the network manipulating the content of that information is provided by the bearer network.
- There is no need for the network to process the information and therefore does not change the content.
- Bearer services belong to the first three layers of the OSI model.
- They are well defined in the ISDN standard. They can be provided using circuit-switched, packet-switched, frame-switched, or cell-switched networks.

Tele Services:

- In this the network may change or process the contents of the data.
- These services corresponds to layers 4-7 of the OSI model.
- Teleservices relay on the facilities of the bearer services and are designed to accommodate complex user needs.
- The user need not to be aware of the details of the process.
- Teleservices include telephony, teletex, telefax, videotex, telex and teleconferencing.
- Though the ISDN defines these services by name yet they have not yet become standards.

Supplementary Services:

- Supplementary services play a vital role in extending the capabilities of both bearer services and teleservices.
- They introduce additional features and options to enhance the overall user experience.
- Some examples of this is call waiting, message handling, call forwarding, call barring ,call transfer etc.



ISDN:

Advantages of ISDN:

- Reliable Connection: Stable digital connection with fewer errors.
- Multiple Digital Channels: Supports simultaneous voice, video, and data.
- Faster Data Transfer: Higher speeds than analog lines.
- Efficient Bandwidth Use: More effective use of network resources.
- Improved Call Quality: Clearer voice calls.
- Integrated Services: Supports voice, video conferencing, fax, and internet.

Disadvantages of ISDN:

- Higher Costs: More expensive than traditional phone systems.
- Specialized Devices: Requires ISDN modems and adapters.
- Less Flexible: Limited speed upgrades and modifications.
- Limited Coverage: Not available in all areas, especially rural regions.
- Obsolescence: Largely replaced by modern broadband technologies.
- Limited Features: Fewer advanced features compared to newer technologies.

ISDN Standards:

- ITU-T groups and organizes the ISDN protocols according to the following general topic areas:
  - E-Protocols: Recommend telephone network standards for ISDN. For examples, international addressing for ISDN.
  - I-Protocols: Deal with concepts, terminology, and general methods.
  - Q-Protocols: Cover how switching and signalling should operate. The term signalling in this context means the process of establishing an ISDN call.

Issue	Protocol	Key Examples
Telephone network and ISDN	E-series	E.163—International telephone numbering plan E.164—International ISDN addressing
ISDN concepts, aspects, and interfaces	I-series	I.100 Series—Concepts, structures, terminology I.400—User-Network Interfaces (UNIs)
Switching and signaling	Q-series	Q.921—LAPD (Link Access Procedure on the D channel) Q.931—ISDN network layer between terminal and switch

Standards from the ITU-T (formerly CCITT)

PYQs:

- What do you understand by circuit switching? Explain. What are its advantage and disadvantage ? **2080**
- What do you understand by packet switching? Explain. What are its advantages and disadvantages? **2080**
- Write Short Notes:
  - ISDN **2078**
  - Message Switching
  - Connection Oriented Services
  - Bridge
- What is transmission media? How do guided media differ from unguided media? Explain different types of guided media in detail. **2078**
- List different types of networking devices and hence point out major differences among hub, switch and router.
- Explain any two wireless transmission media. **2081**
- What is switching? Compare and contrast a circuit-switched network and packet-switched network.

Assignment-2:

- What do you understand by circuit switching? Explain. What are its advantage and disadvantage ? 2080
- What do you understand by packet switching? Explain. What are its advantages and disadvantages? 2080
- Write Short Notes:
    - ISDN 2078
    - Message Switching
    - Connection Oriented Services
    - Bridge
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THANKS!

❓ **Do you have any questions?**

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